

CS 57800 – Statistical Machine Learning

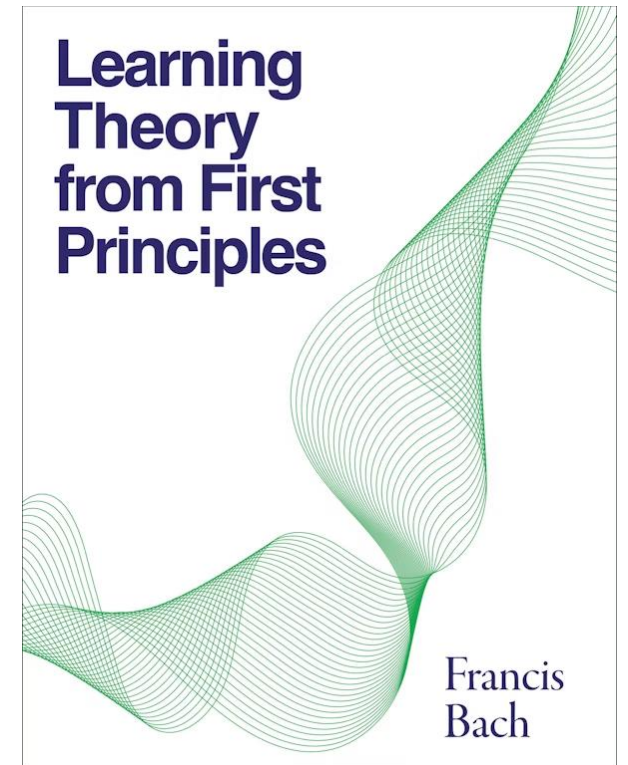
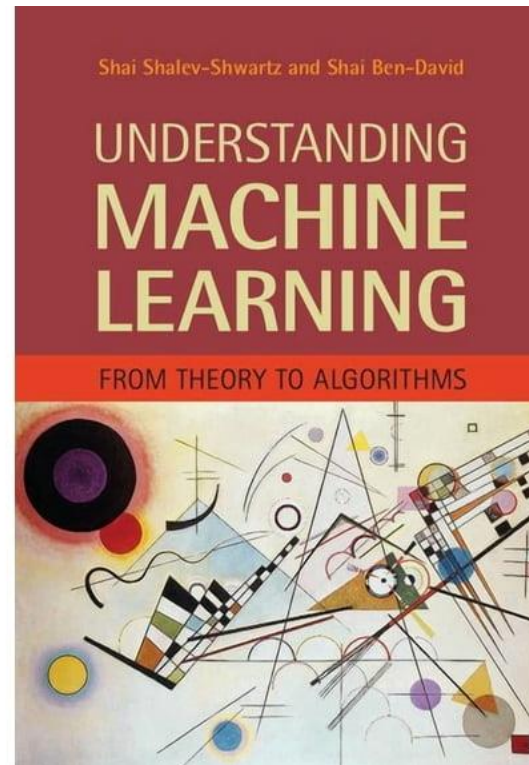
Lecture 1 Learning Theory

https://ruizhezhang.com/course_fall_2026.html

Outline

Intro to Learning Theory

- Statistical Learning Framework
- PAC Learning
- The VC Dimension



The Statistical Learning Framework

- **Domain set $\mathcal{X}$:** images, sounds, text, DNA sequences, web pages, ...
- **Label set $\mathcal{Y}$:** for binary classification, $\mathcal{Y} = \{0,1\}$ or $\{-1,1\}$; for multcategory classification, $\mathcal{Y} = \{1,2, \dots, k\}$; and for classical regression, $\mathcal{Y} = \mathbb{R}$
- **Training data $S = ((x_1, y_1), \dots, (x_m, y_m)) \in (\mathcal{X} \times \mathcal{Y})^m$**
- **Data-generation model:** a probability distribution $\mathcal{D}$ over $\mathcal{X} \times \mathcal{Y}$
 - Note that y_i may not be a deterministic function of x_i . E.g., $y = f(x) + \varepsilon$ with $\mathbb{E}[\varepsilon] = 0$, or $y = f(x, z)$ with unobserved z
 - $\mathcal{D}$ is unknown to the learner
- **Machine learning algorithm A :** a function that maps the training data S to a function (prediction rule/predictor/classifier/hypothesis) from $\mathcal{X}$ to $\mathcal{Y}$



The Statistical Learning Framework

- Loss function $\ell : \mathcal{Y} \times \mathcal{Y} \rightarrow \mathbb{R}$
 - 0-1 loss $\ell(y, z) = \mathbf{1}\{y \neq z\}$
 - square loss $\ell(y, z) = (y - z)^2$
- Expected risk / Population loss / Generalization error / Test error of f :

$$\mathcal{R}(f) := \mathbb{E}_{(x,y) \sim \mathcal{D}}[\ell(f(x), y)]$$

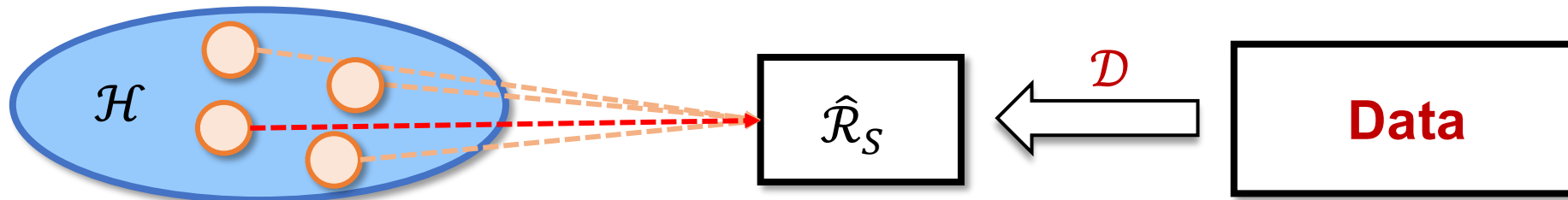
- Empirical risk / Empirical loss / Training error of f :

$$\hat{\mathcal{R}}_S(f) := \frac{1}{m} \sum_{i=1}^m \ell(f(x_i), y_i)$$

Empirical Risk Minimization

- Let $\mathcal{H}$ be a hypothesis class $\{f : \mathcal{X} \rightarrow \mathcal{Y}\}$
- An ERM learner aims at finding a function $f \in \mathcal{H}$ that minimizes the empirical risk:

$$\text{ERM}_{\mathcal{H}}(S) \in \arg \min_{f \in \mathcal{H}} \hat{R}_S(f)$$



Example: ordinary linear regression

- $\mathcal{H} = \{f_{\theta}(x) = \theta^{\top} \phi(x)\}$ where $\phi(\cdot)$ is a fixed feature map $\mathcal{X} \rightarrow \mathbb{R}^d$
- ERM is the least squares minimization:

$$\min_{\theta} \frac{1}{m} \sum_{i=1}^m (\theta^{\top} \phi(x_i) - y_i)^2$$

Empirical Risk Minimization

The Bayes optimal predictor:

$$\begin{aligned}\mathcal{R}(f) &= \mathbb{E}_{(x,y) \sim \mathcal{D}}[\ell(f(x), y)] = \mathbb{E}[\mathbb{E}[\ell(f(x), y) | x]] \\ &= \int_{\mathcal{X}} \underbrace{\mathbb{E}[\ell(f(x'), y) | x = x']}_{\text{conditional risk}} dp(x'),\end{aligned}$$

where p is the density of the marginal distribution of $\mathcal{D}$ on $\mathcal{X}$

- To minimize the expected risk $\mathcal{R}(f)$, we just need to minimize the conditional risk for each $x' \in \mathcal{X}$

$$f_{\star}(x') := \arg \min_{z \in \mathcal{Y}} \mathbb{E}[\ell(z, y) | x = x'], \quad \forall x' \in \mathcal{X}$$

- The Bayes risk is

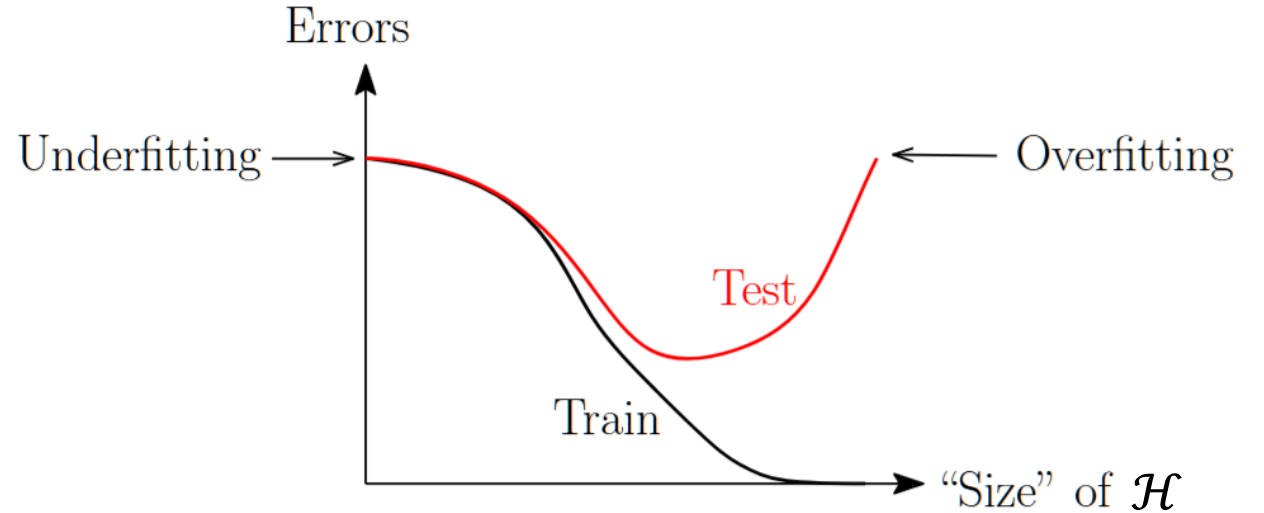
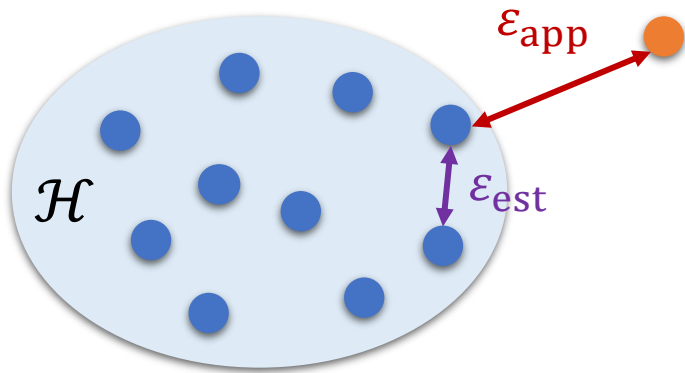
$$\mathcal{R}_{\star} := \mathcal{R}(f_{\star}) = \int_{\mathcal{X}} \min_{z \in \mathcal{Y}} \mathbb{E}[\ell(z, y) | x = x'] dp(x')$$

- **Exercise:** Show that the Bayes optimal predictor $f_{\star}$ is optimal, i.e., $\mathcal{R}_{\star} \leq \mathcal{R}(f)$ for any $f: \mathcal{X} \rightarrow \mathcal{Y}$

Empirical Risk Minimization

Risk decomposition:

$$R(f) - \mathcal{R}_\star = \underbrace{\min_{h \in \mathcal{H}} R(h) - \mathcal{R}_\star}_{\text{approximation error}} + \underbrace{R(f) - \min_{h \in \mathcal{H}} R(h)}_{\text{estimation error}}$$



Toy Example: Polynomial Regression

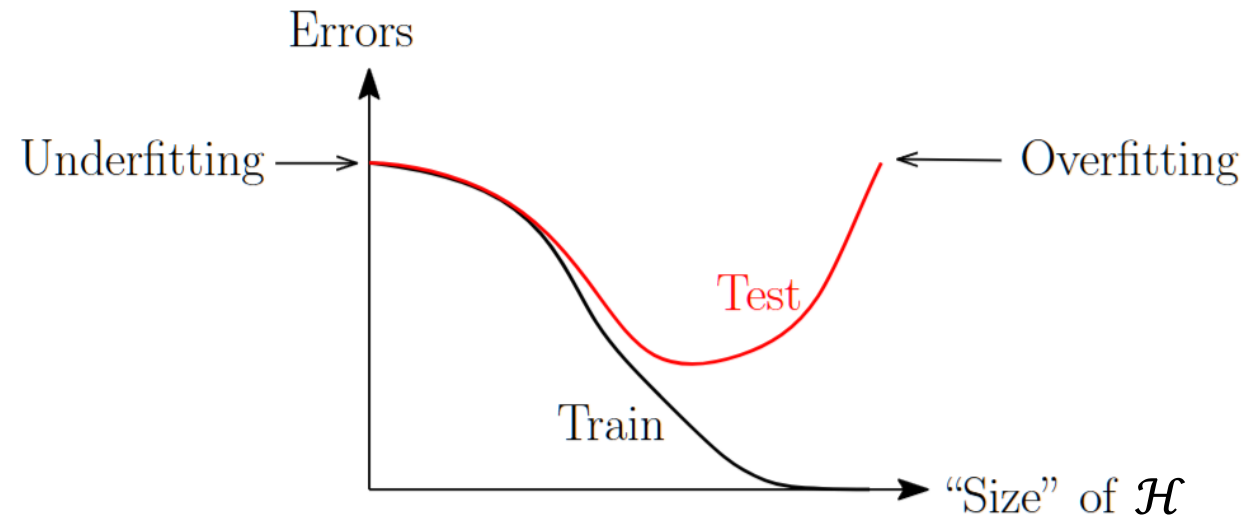
The goal is to find the best degree- P polynomial approximation of the function

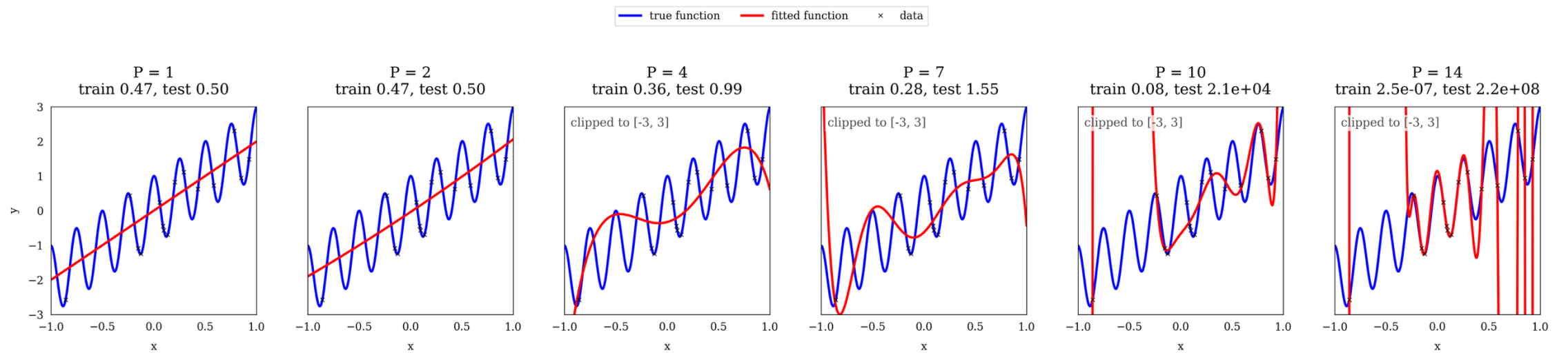
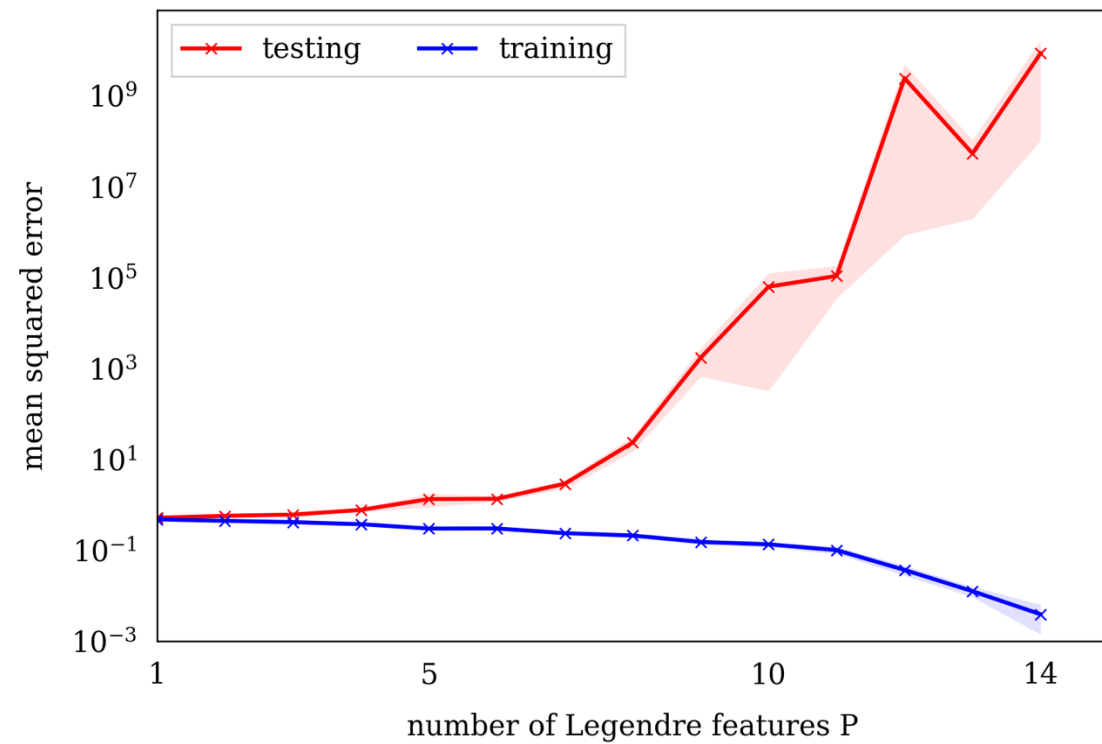
$$f(x) = 2x + \cos(25x)$$

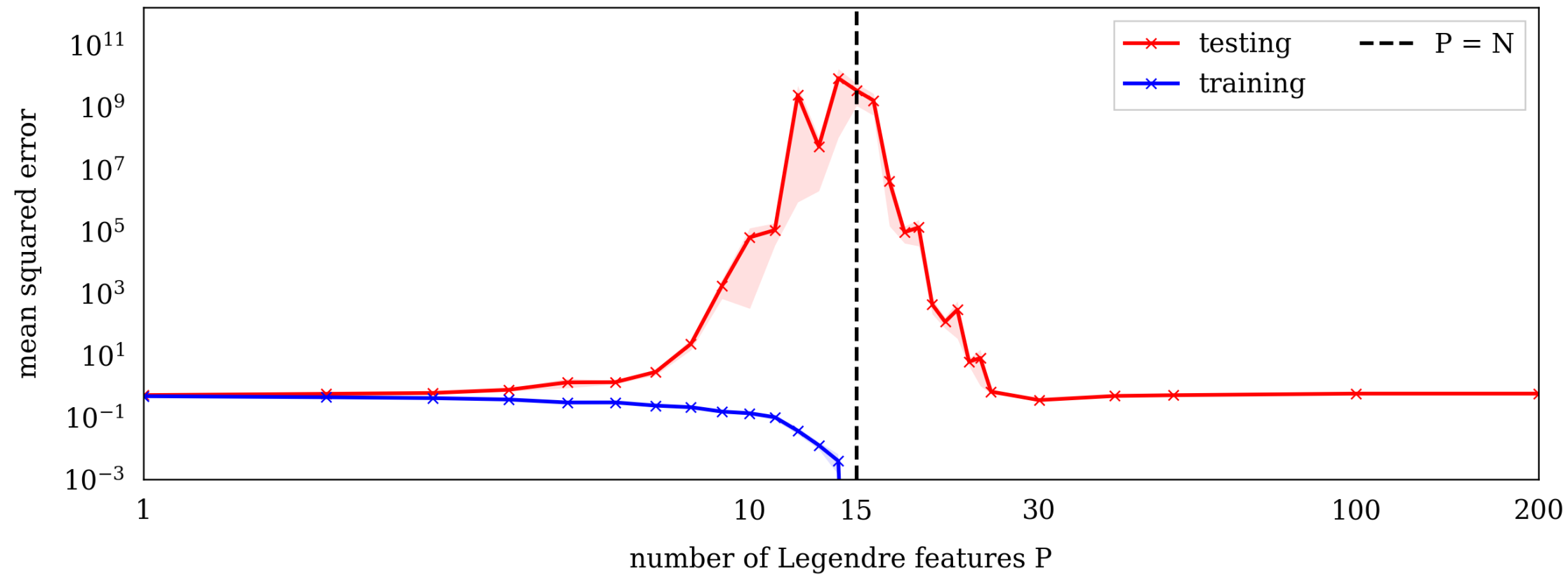
- Training data: $\{(x_i, f(x_i))\}_{i \leq N}$, each $x_i \in [-1, 1]$ is an i.i.d. uniform sample
- Hypothesis class:

$$\mathcal{H} = \{\theta \in \mathbb{R}^P : \theta^\top \phi(x)\}, \quad \phi(x) = (L_1(x), L_2(x), \dots, L_P(x))$$

Legendre polynomials







(Schaeffer et al., arXiv:2303.14151)

PAC Learning

Probably approximately correct (PAC) learning

A hypothesis class $\mathcal{H}$ is **(agnostic) PAC learnable** if there exist a function $m_{\mathcal{H}}$ and a learning algorithm A such that for any $\delta \in (0, 1)$ and $\varepsilon > 0$, for any distribution $\mathcal{D}$ over $\mathcal{X} \times \mathcal{Y}$, when running the learning algorithm on $m \geq m_{\mathcal{H}}(\varepsilon, \delta)$ i.i.d. samples S generated by $\mathcal{D}$, the algorithm returns a hypothesis $A(S) : \mathcal{X} \rightarrow \mathcal{Y}$ satisfying

$$\Pr_{S \sim \mathcal{D}^m} \left[\mathcal{R}(A(S)) - \inf_{f \in \mathcal{H}} \mathcal{R}(f) \leq \varepsilon \right] \geq 1 - \delta$$

- Today's PAC learning definition is slightly different from the original one; see [\(Hanneke-Mehrotra-Velegkas-Zampetakis, arXiv:2605.13840\)](#)
- PAC learning has strong connections to many subfields of theoretical computer science apart from learning, including property testing, algorithmic game theory, computational complexity, statistical algorithms, and cryptography; see [\(Servedio, arXiv:2511.08791\)](#)



Leslie Valiant
Turing Award Laureate

PAC Learning

Given a hypothesis class $\mathcal{H}$, is it PAC learnable? How many samples do we need?

The Realizability Assumption: there exists an $h \in \mathcal{H}$ such that for any $(x, y) \sim \mathcal{D}$ satisfies $y = h(x)$. In other words, $\mathcal{R}(h) = 0$

Proposition. Every **finite** hypothesis class is PAC learnable with sampling complexity

$$m(\varepsilon, \delta) \leq O\left(\frac{\log(|\mathcal{H}|/\delta)}{\varepsilon}\right)$$

- There exists a learner A such that for any $m \geq \varepsilon^{-1} \log(|\mathcal{H}|/\delta)$,
$$\Pr_{S \sim \mathcal{D}^m}[\mathcal{R}(A(S)) \leq \varepsilon] \geq 1 - \delta$$

- With high probability over $S \sim \mathcal{D}^m$, the empirical risk minimization satisfies

Generalization bound

$$\mathcal{R}(\text{ERM}_{\mathcal{H}}(S)) - \underbrace{\hat{\mathcal{R}}_S(\text{ERM}_{\mathcal{H}}(S))}_{= 0} \leq O\left(\frac{\log(|\mathcal{H}|)}{m}\right)$$

PAC Learning

Proposition. Every finite hypothesis class is PAC learnable with sampling complexity

$$m(\varepsilon, \delta) \leq O\left(\frac{\log(|\mathcal{H}|/\delta)}{\varepsilon}\right)$$

Proof.

- It suffices to prove that, for $m \geq \varepsilon^{-1} \log(|\mathcal{H}|/\delta)$, $\Pr_{S \sim \mathcal{D}^m}[\mathcal{R}(\text{ERM}_{\mathcal{H}}(S)) > \varepsilon] \leq \delta$

- Let $\mathcal{H}_B := \{f \in \mathcal{H} : \mathcal{R}(f) > \varepsilon\}$ be the set of “bad” hypothesis. That is,

$$\Pr_{S \sim \mathcal{D}^m}[\mathcal{R}(\text{ERM}_{\mathcal{H}}(S)) > \varepsilon] = \Pr_{S \sim \mathcal{D}^m}[\text{ERM}_{\mathcal{H}}(S) \in \mathcal{H}_B]$$

- Let $\mathcal{M} := \{S : \exists f \in \mathcal{H}_B, \hat{\mathcal{R}}_S(f) = 0\}$ be the set of “misleading” datasets. Then, we have

$$\Pr_{S \sim \mathcal{D}^m}[\text{ERM}_{\mathcal{H}}(S) \in \mathcal{H}_B] \leq \Pr_{S \sim \mathcal{D}^m}[S \in \mathcal{M}] = \Pr_{S \sim \mathcal{D}^m}[\exists f \in \mathcal{H}_B, \hat{\mathcal{R}}_S(f) = 0]$$

PAC Learning

Proposition. Every finite hypothesis class is PAC learnable with sampling complexity

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(Union bound)

PAC Learning

Proposition. Every finite hypothesis class is PAC learnable with sampling complexity

$$m(\varepsilon, \delta) \leq O\left(\frac{\log(|\mathcal{H}|/\delta)}{\varepsilon}\right)$$

Proof.

$$\Pr_{S \sim \mathcal{D}^m}[\text{ERM}_{\mathcal{H}}(S) \in \mathcal{H}_B] \leq \sum_{f \in \mathcal{H}_B} \Pr_{S \sim \mathcal{D}^m}[\hat{\mathcal{R}}_S(f) = 0] = \sum_{f \in \mathcal{H}_B} \Pr_{S \sim \mathcal{D}^m}[\forall i \in [m], f(x_i) = y_i]$$

ℓ is the 0-1 loss or bounded loss



PAC Learning

Proposition. Every finite hypothesis class is PAC learnable with sampling complexity

$$m(\varepsilon, \delta) \leq O\left(\frac{\log(|\mathcal{H}|/\delta)}{\varepsilon}\right)$$

Proof.

$$\begin{aligned}\Pr_{S \sim \mathcal{D}^m}[\text{ERM}_{\mathcal{H}}(S) \in \mathcal{H}_B] &\leq \sum_{f \in \mathcal{H}_B} \Pr_{S \sim \mathcal{D}^m}[\hat{\mathcal{R}}_S(f) = 0] = \sum_{f \in \mathcal{H}_B} \Pr_{S \sim \mathcal{D}^m}[\forall i \in [m], f(x_i) = y_i] \\ &= \sum_{f \in \mathcal{H}_B} \left(\Pr_{(x,y) \sim \mathcal{D}}[f(x) = y] \right)^m \\ &\leq \sum_{f \in \mathcal{H}_B} \left(\mathbb{E}_{(x,y) \sim \mathcal{D}}[1 - \ell(f(x), y)] \right)^m \quad \ell \text{ is the 0-1 loss or bounded loss} \\ &= \sum_{f \in \mathcal{H}_B} (1 - \mathcal{R}(f))^m \leq \sum_{f \in \mathcal{H}_B} (1 - \varepsilon)^m \leq |\mathcal{H}_B| e^{-\varepsilon m} \leq \delta\end{aligned}$$


PAC Learning

“No Free Lunch” Theorem

Consider binary classification with 0–1 loss and $\mathcal{X}$ infinite. For any $m > 0$ and any learning algorithm A , there exists a probability distribution $\mathcal{D}$ such that

$$\mathcal{R}_\star = 0 \quad \text{and} \quad \mathbb{E}_{S \sim \mathcal{D}^m} [\mathcal{R}(A(S))] \geq 1/4$$

- **No universal learner:** for every learner, there exists a task on which it fails, even though that task can be successfully learned by another learner
- **Exercise:** Let $\mathcal{X}$ be infinite and let $\mathcal{H}$ be the set of **all functions** from $\mathcal{X}$ to $\{0,1\}$. Then, $\mathcal{H}$ is not PAC learnable

PAC Learning

“No Free Lunch” Theorem Consider binary classification with 0–1 loss and $\mathcal{X}$ infinite. For any $m > 0$ and any learning algorithm A , there exists a probability distribution $\mathcal{D}$ such that

$$\mathcal{R}_\star = 0 \quad \text{and} \quad \mathbb{E}_{S \sim \mathcal{D}^m}[\mathcal{R}(A(S))] \geq 1/4$$

Proof.

- Fix any $m > 0$. The idea is to construct a hard distribution $\mathcal{D}$ supported on $k \gg m$ data points such that the knowledge of the m training data point is not enough to predict the remaining $k - m$ data points
- Pick any k different elements $x_1, \dots, x_k \in \mathcal{X}$
- For every $r \in \{0,1\}^k$, define a probability distribution $\mathcal{D}_r$:

$$\mathcal{D}_r((x_i, y)) := \begin{cases} 1/k & \text{if } y = r_i \\ 0 & \text{otherwise} \end{cases}, \quad i \in [k], y \in \{0,1\}$$

- $\mathcal{R}_\star = 0$ for every $\mathcal{D}_r$

PAC Learning

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$$\mathcal{R}_\star = 0 \quad \text{and} \quad \mathbb{E}_{S \sim \mathcal{D}^m} [\mathcal{R}(A(S))] \geq 1/4$$

Proof.

- Probabilistic method:

$$\begin{aligned} \max_{r \in \{0,1\}^k} \{ \mathbb{E}_{S \sim \mathcal{D}_r^m} [\mathcal{R}(A(S))] \} &\geq \mathbb{E}_{r \sim \{0,1\}^k} \left[\mathbb{E}_{S \sim \mathcal{D}_r^m} [\mathcal{R}(A(S))] \right] \\ &= \mathbb{E}_{r,S} \left[\Pr_{i \sim [k]} [A(S)(x_i) \neq r_i] \right] = \Pr_{r,S,i} [A(S)(x_i) \neq r_i] \end{aligned}$$

- To prove the existence of some object with certain property, we devise a *random construction* and show that it works with *positive probability*. (Here, we use $\Pr[X \geq \mathbb{E}[X]] > 0$)

PAC Learning

“No Free Lunch” Theorem Consider binary classification with 0–1 loss and $\mathcal{X}$ infinite. For any $m > 0$ and any learning algorithm A , there exists a probability distribution $\mathcal{D}$ such that

$$\mathcal{R}_\star = 0 \quad \text{and} \quad \mathbb{E}_{S \sim \mathcal{D}^m}[\mathcal{R}(A(S))] \geq 1/4$$

Proof.

- Let $S = (x_{i_1}, r_{i_1}, \dots, x_{i_m}, r_{i_m})$

$$\Pr_{r, S, i}[A(S)(x_i) \neq r_i] = \mathbb{E} \left[\Pr_{r, i}[A(S)(x_i) \neq r_i \mid x_{i_1}, r_{i_1}, \dots, x_{i_m}, r_{i_m}] \right]$$

PAC Learning

“No Free Lunch” Theorem Consider binary classification with 0–1 loss and $\mathcal{X}$ infinite. For any $m > 0$ and any learning algorithm A , there exists a probability distribution $\mathcal{D}$ such that

$$\mathcal{R}_\star = 0 \quad \text{and} \quad \mathbb{E}_{S \sim \mathcal{D}^m}[\mathcal{R}(A(S))] \geq 1/4$$

Proof.

- Let $S = (x_{i_1}, r_{i_1}, \dots, x_{i_m}, r_{i_m})$

$$\begin{aligned} \Pr_{r,S,i}[A(S)(x_i) \neq r_i] &= \mathbb{E} \left[\Pr_{r,i}[A(S)(x_i) \neq r_i \mid x_{i_1}, r_{i_1}, \dots, x_{i_m}, r_{i_m}] \right] \\ &\geq \mathbb{E} \left[\Pr_{r,i}[A(S)(x_i) \neq r_i \ \& \ x_i \notin \{x_{i_1}, \dots, x_{i_m}\} \mid x_{i_1}, r_{i_1}, \dots, x_{i_m}, r_{i_m}] \right] \\ &= \mathbb{E} \left[\frac{1}{2} \cdot \Pr_i[x_i \notin \{x_{i_1}, \dots, x_{i_m}\} \mid x_{i_1}, r_{i_1}, \dots, x_{i_m}, r_{i_m}] \right] \\ &= \frac{1}{2} \mathbb{E} \left[1 - \frac{|\{x_{i_1}, \dots, x_{i_m}\}|}{k} \right] \geq \frac{1}{2} \left(1 - \frac{m}{k} \right) \end{aligned}$$

PAC Learning

“No Free Lunch” Theorem Consider binary classification with 0–1 loss and $\mathcal{X}$ infinite. For any $m > 0$ and any learning algorithm A , there exists a probability distribution $\mathcal{D}$ such that

$$\mathcal{R}_\star = 0 \quad \text{and} \quad \mathbb{E}_{S \sim \mathcal{D}^m} [\mathcal{R}(A(S))] \geq 1/4$$

Proof.

- Thus, we get that

$$\max_{r \in \{0,1\}^k} \{ \mathbb{E}_{S \sim \mathcal{D}_r^m} [\mathcal{R}(A(S))] \} \geq \frac{1}{2} \left(1 - \frac{m}{k} \right) \geq \frac{1}{4}$$

when $k \geq 2m$



PAC Learning

- We have shown that under the **realizability assumption** ($\exists h \in \mathcal{H}, \mathcal{R}(h) = 0$), every finite $\mathcal{H}$ is PAC learnable
- We can remove this assumption:

Theorem. Let $\mathcal{H}$ be a finite class, and ℓ be the loss function bounded in $[0,1]$. Then, $\mathcal{H}$ is **agnostic** PAC learnable using the ERM algorithm with sample complexity

$$m(\varepsilon, \delta) \leq O\left(\frac{\log(|\mathcal{H}|/\delta)}{\varepsilon^2}\right)$$

- A byproduct is the **Uniform Convergence property**: for any $\mathcal{D}$, with high probability over $S \sim \mathcal{D}^m$,

$$\max_{f \in \mathcal{H}} |\mathcal{R}(f) - \hat{\mathcal{R}}_S(f)| \leq O\left(\sqrt{\frac{\log(|\mathcal{H}|)}{m}}\right)$$

PAC Learning

Theorem. Let $\mathcal{H}$ be a finite class, and ℓ be the loss function bounded in $[0,1]$. Then, $\mathcal{H}$ is agnostically PAC learnable using the ERM algorithm with sample complexity

$$m(\varepsilon, \delta) \leq O\left(\frac{\log(|\mathcal{H}|/\delta)}{\varepsilon^2}\right)$$

Proof.

- First, we want to prove that for $m \geq m(\varepsilon, \delta)$,

$$\Pr_{S \sim \mathcal{D}^m} \left[\mathcal{R}(\text{ERM}_{\mathcal{H}}(S)) - \inf_{f \in \mathcal{H}} \mathcal{R}(f) \geq \varepsilon \right] \leq \delta$$

- Let $h := \arg \min_{f \in \mathcal{H}} \mathcal{R}(f)$. We have the following error decomposition:

$$\begin{aligned} & \mathcal{R}(\text{ERM}_{\mathcal{H}}(S)) - \mathcal{R}(h) \\ &= \left(\mathcal{R}(\text{ERM}_{\mathcal{H}}(S)) - \hat{\mathcal{R}}_S(\text{ERM}_{\mathcal{H}}(S)) \right) + \overbrace{\left(\hat{\mathcal{R}}_S(\text{ERM}_{\mathcal{H}}(S)) - \hat{\mathcal{R}}_S(h) \right)}^{\leq 0} + \left(\hat{\mathcal{R}}_S(h) - \mathcal{R}(h) \right) \\ &\leq \left(\mathcal{R}(\text{ERM}_{\mathcal{H}}(S)) - \hat{\mathcal{R}}_S(\text{ERM}_{\mathcal{H}}(S)) \right) + \left(\hat{\mathcal{R}}_S(h) - \mathcal{R}(h) \right) \end{aligned}$$

PAC Learning

Theorem. Let $\mathcal{H}$ be a finite class, and ℓ be the loss function bounded in $[0,1]$. Then, $\mathcal{H}$ is agnostically PAC learnable using the ERM algorithm with sample complexity

$$m(\varepsilon, \delta) \leq O\left(\frac{\log(|\mathcal{H}|/\delta)}{\varepsilon^2}\right)$$

Proof.

$$\begin{aligned}\Pr_{S \sim \mathcal{D}^m}[\mathcal{R}(\text{ERM}_{\mathcal{H}}(S)) - \mathcal{R}(h) \geq \varepsilon] &\leq \Pr_{S \sim \mathcal{D}^m}\left[\left(\mathcal{R}(\text{ERM}_{\mathcal{H}}(S)) - \hat{\mathcal{R}}_S(\text{ERM}_{\mathcal{H}}(S))\right) + \left(\hat{\mathcal{R}}_S(h) - \mathcal{R}(h)\right) \geq \varepsilon\right] \\ &\leq \Pr_{S \sim \mathcal{D}^m}[\exists f \in \mathcal{H}, |\mathcal{R}(f) - \hat{\mathcal{R}}_S(f)| \geq \varepsilon/2] \\ &\leq \sum_{f \in \mathcal{H}} \Pr_{S \sim \mathcal{D}^m}[|\mathcal{R}(f) - \hat{\mathcal{R}}_S(f)| \geq \varepsilon/2]\end{aligned}$$

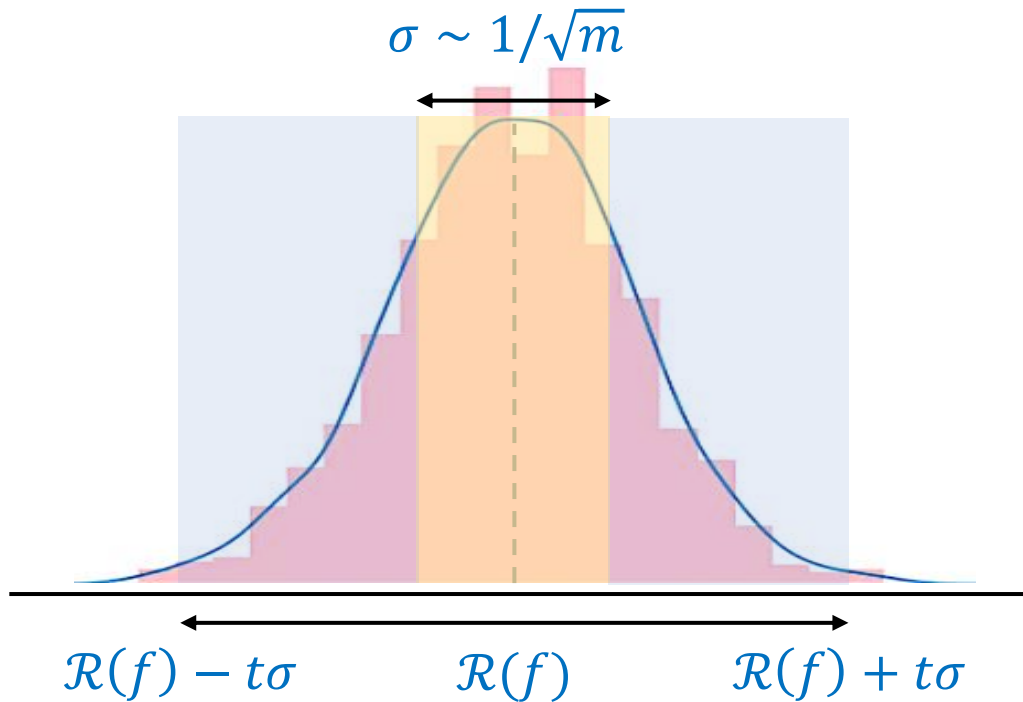
Random variable

- Note that

$$\hat{\mathcal{R}}_S(f) = \frac{1}{m} \sum_{i=1}^m \ell(f(x_i), y_i), \quad \mathcal{R}(f) = \mathbb{E}_{(x,y) \sim \mathcal{D}}[\ell(f(x), y)], \quad \mathbb{E}[\hat{\mathcal{R}}(f)] = \mathcal{R}(f)$$

Detour: Concentration

Central limit theorem / Hoeffding / Chernoff / ...



$$\Pr[|X - \mathbb{E}[X]| \geq t\sigma] \leq \exp(-\Omega(t^2))$$

Hoeffding's inequality

Let $X_1, \dots, X_m$ be i.i.d. random variables with $\mathbb{E}[X_i] = \mu$ and $a \leq X_i \leq b$. Then,

$$\Pr\left[\left|\frac{1}{m} \sum_{i=1}^m X_i - \mu\right| \geq \epsilon\right] \leq 2 \exp\left(-\frac{2m\epsilon^2}{(b-a)^2}\right)$$

https://ruizhezhang.com/teaching_2026/Lecture%203.pdf

PAC Learning

Theorem. Let $\mathcal{H}$ be a finite class, and ℓ be the loss function bounded in $[0,1]$. Then, $\mathcal{H}$ is agnostically PAC learnable using the ERM algorithm with sample complexity

$$m(\varepsilon, \delta) \leq O\left(\frac{\log(|\mathcal{H}|/\delta)}{\varepsilon^2}\right)$$

Proof.

- By Hoeffding's inequality,

$$\Pr_{S \sim \mathcal{D}^m} [|\mathcal{R}(f) - \hat{\mathcal{R}}_S(f)| \geq \varepsilon/2] \leq \exp(-\Omega(m\varepsilon^2))$$

- Thus,

$$\Pr_{S \sim \mathcal{D}^m} \left[\mathcal{R}(\text{ERM}_{\mathcal{H}}(S)) - \inf_{f \in \mathcal{H}} \mathcal{R}(f) \geq \varepsilon \right] \leq |\mathcal{H}| \cdot \exp(-\Omega(m\varepsilon^2)) \leq \delta$$

as long as

$$m \geq \Omega\left(\frac{\log(|\mathcal{H}|/\delta)}{\varepsilon^2}\right)$$



PAC Learning

Theorem. Let $\mathcal{H}$ be a finite class, and ℓ be the loss function bounded in $[0,1]$. Then, $\mathcal{H}$ is agnostically PAC learnable using the ERM algorithm with sample complexity

$$m(\varepsilon, \delta) \leq O\left(\frac{\log(|\mathcal{H}|/\delta)}{\varepsilon^2}\right)$$

What about infinite class $|\mathcal{H}| = \infty$?

- **No Free Lunch theorem** tells that $\mathcal{H}$ cannot be all functions from an infinite domain $\mathcal{X}$ to $\{0,1\}$
- **Exercise:** Let $\mathcal{X} = \mathbb{R}$ and $\mathcal{H}$ be the set of indicator functions of closed intervals:

$$\mathbf{1}\{a \leq x \leq b\}, \quad \forall a \leq b$$

Then $\mathcal{H}$ can be PAC-learned using ERM with sample complexity $O\left(\frac{1}{\varepsilon} \log \frac{1}{\delta}\right)$

The VC Dimension

Shattering: A hypothesis class $\mathcal{H}$ shatters a finite set $C \subset \mathcal{X}$ if the restriction of $\mathcal{H}$ to C is the set of all functions from C to $\{0, 1\}$, i.e., for $C = \{c_1, \dots, c_m\}$,

$$\mathcal{H}_C := \{(h(c_1), \dots, h(c_m)) : h \in \mathcal{H}\} = \{(y_1, \dots, y_m) : y_i \in \{0, 1\}\}$$

That is, $|\mathcal{H}_C| = 2^{|C|}$

	c_1	c_2	c_3	c_4	c_5
h_1	0	0	0	0	0
h_2	1	1	1	1	1
h_3	1	1	1	0	0
h_4	0	0	0	1	1

The VC Dimension

Shattering: A hypothesis class $\mathcal{H}$ shatters a finite set $C \subset \mathcal{X}$ if the restriction of $\mathcal{H}$ to C is the set of all functions from C to $\{0, 1\}$, i.e., for $C = \{c_1, \dots, c_m\}$,

$$\mathcal{H}_C := \{(h(c_1), \dots, h(c_m)) : h \in \mathcal{H}\} = \{(y_1, \dots, y_m) : y_i \in \{0, 1\}\}$$

That is, $|\mathcal{H}_C| = 2^{|C|}$

VC-dimension

The VC-dimension of a hypothesis class $\mathcal{H}$, denoted $\text{VCdim}(\mathcal{H})$, is the maximal size of a set $C \subset \mathcal{X}$ that can be shattered by $\mathcal{H}$. If $\mathcal{H}$ can shatter sets of arbitrarily large size, we say $\text{VCdim}(\mathcal{H}) = \infty$

- If $\text{VCdim}(\mathcal{H}) = \infty$, then $\mathcal{H}$ is not PAC learnable



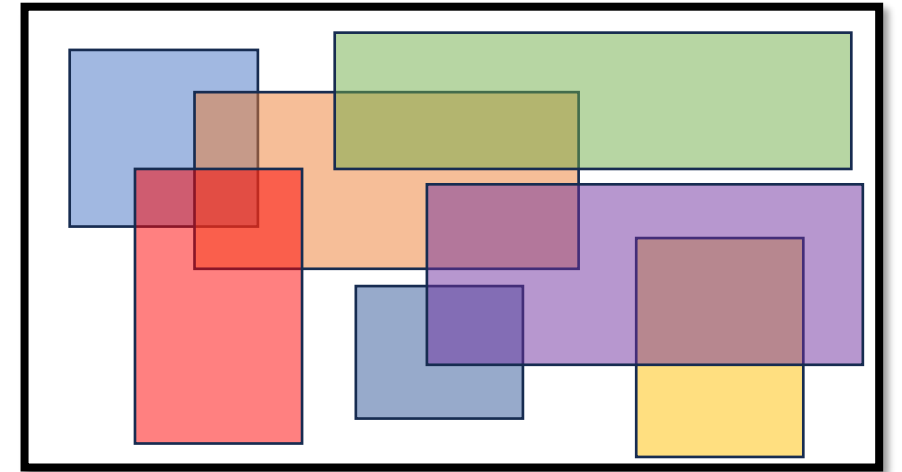
Alexei Chervonenkis and Vladimir Vapnik

The VC Dimension

The VC-dimension of a hypothesis class $\mathcal{H}$, denoted $\text{VCdim}(\mathcal{H})$, is the **maximal** size of a set $C \subset \mathcal{X}$ that can be shattered by $\mathcal{H}$.

Examples:

- Finite $\mathcal{H}$: $2^{|C|} = |\mathcal{H}_C| \leq |\mathcal{H}| \Rightarrow \text{VCdim}(\mathcal{H}) \leq \log|\mathcal{H}|$
- Axis aligned rectangles: $\text{VCdim}(\mathcal{H}) = 4$



The VC Dimension

The Fundamental Theorem of Statistical Learning.

Let $\mathcal{H}$ be a hypothesis class of functions from $\mathcal{X}$ to $\{0, 1\}$ and let the loss function be the 0–1 loss. Then, the following are equivalent:

- 1) $\mathcal{H}$ is agnostic PAC learnable
- 2) $\mathcal{H}$ is PAC learnable
- 3) $\mathcal{H}$ has finite VC dimension

- We already have $1) \Rightarrow 2)$ and $2) \Rightarrow 3)$. It remains to show that $3) \Rightarrow 1)$
- It is very similar to the agnostic PAC learning theorem for finite $\mathcal{H}$:

Let $\mathcal{H}$ be a finite class. Then, $\mathcal{H}$ is agnostic PAC learnable with sample complexity $O\left(\frac{\log(|\mathcal{H}|/\delta)}{\varepsilon^2}\right)$

- The VC dimension bounds the “effective size” of $\mathcal{H}$

The VC Dimension

Lemma (Sauer-Shelah-Perles). Let $\mathcal{H}$ be a hypothesis class with $\text{VCdim}(\mathcal{H}) \leq d < \infty$. Then, for all m ,

$$\tau_{\mathcal{H}}(m) := \max_{C \subset \mathcal{X}: |C| \leq m} |\mathcal{H}_C| \leq \sum_{i=0}^d \binom{m}{i}$$

In particular, if $m \geq d + 1$, then $\tau_{\mathcal{H}}(m) \leq (em/d)^d$

- For any subset C , the “effective size” $|\mathcal{H}_C| = O(|C|^d)$ instead of $2^{|C|}$
- We only prove the uniform convergence property, which will give agnostic PAC learnability

Proposition (Uniform Convergence). Let $\mathcal{H}$ be a class and let $\tau_{\mathcal{H}}$ be its growth function. Then, for every $\mathcal{D}$, with probability of at least $1 - \delta$ over the choice of $S \sim \mathcal{D}^m$ we have

$$\max_{f \in \mathcal{H}} |\mathcal{R}(f) - \hat{\mathcal{R}}_S(f)| \leq \frac{\sqrt{\log \tau_{\mathcal{H}}(2m)}}{\delta \sqrt{m}}$$

The VC Dimension

Proposition. Let $\mathcal{H}$ be a class and let $\tau_{\mathcal{H}}$ be its growth function. Then, for every $\mathcal{D}$, with probability of at least $1 - \delta$ over the choice of $S \sim \mathcal{D}^m$ we have

$$\max_{f \in \mathcal{H}} |\mathcal{R}(f) - \hat{\mathcal{R}}_S(f)| \leq O\left(\frac{\sqrt{1 + \log \tau_{\mathcal{H}}(2m)}}{\delta \sqrt{m}}\right)$$

Proof.

- By Markov's inequality, it suffices to prove: $\mathbb{E}_S \left[\max_{f \in \mathcal{H}} |\mathcal{R}(f) - \hat{\mathcal{R}}_S(f)| \right] \leq O\left(\sqrt{\frac{1 + \log \tau_{\mathcal{H}}(2m)}{m}}\right)$
- Since $\mathbb{E}_S[\hat{\mathcal{R}}_S(f)] = \mathcal{R}(f)$, we have

$$\begin{aligned} \mathbb{E}_S \left[\max_{f \in \mathcal{H}} |\mathcal{R}(f) - \hat{\mathcal{R}}_S(f)| \right] &= \mathbb{E}_S \left[\max_{f \in \mathcal{H}} |\mathbb{E}_{S'}[\hat{\mathcal{R}}_{S'}(f)] - \hat{\mathcal{R}}_S(f)| \right] \\ &\leq \mathbb{E}_S \left[\max_{f \in \mathcal{H}} \{ \mathbb{E}_{S'} [|\hat{\mathcal{R}}_{S'}(f) - \hat{\mathcal{R}}_S(f)|] \} \right] \end{aligned}$$

The VC Dimension

- By Jensen's inequality,

$$\max_{f \in \mathcal{H}} \left\{ \mathbb{E}_{S'} [|\hat{\mathcal{R}}_{S'}(f) - \hat{\mathcal{R}}_S(f)|] \right\} \leq \mathbb{E}_{S'} \left[\max_{f \in \mathcal{H}} |\hat{\mathcal{R}}_{S'}(f) - \hat{\mathcal{R}}_S(f)| \right]$$

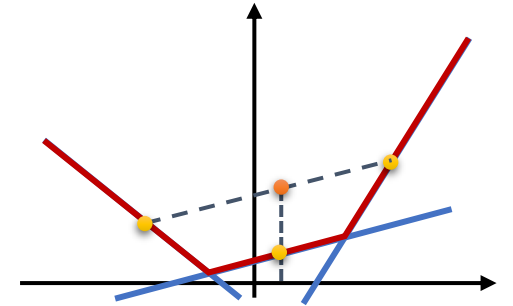
- Thus,

$$\mathbb{E}_S \left[\max_{f \in \mathcal{H}} |\mathcal{R}(f) - \hat{\mathcal{R}}_S(f)| \right] \leq \mathbb{E}_{S,S'} \left[\max_{f \in \mathcal{H}} |\hat{\mathcal{R}}_{S'}(f) - \hat{\mathcal{R}}_S(f)| \right]$$

- Fix $S, S' \sim \mathcal{D}^m$, and let $\mathcal{C} := S \cup S'$ (only the $\mathcal{X}$ part). Then,

$$\max_{f \in \mathcal{H}} |\hat{\mathcal{R}}_{S'}(f) - \hat{\mathcal{R}}_S(f)| = \max_{f \in \mathcal{H}_{\mathcal{C}}} |\hat{\mathcal{R}}_{S'}(f) - \hat{\mathcal{R}}_S(f)|$$

$$\begin{aligned} \mathbb{E}_S \left[\max_{f \in \mathcal{H}} |\mathcal{R}(f) - \hat{\mathcal{R}}_S(f)| \right] &\leq \mathbb{E}_{S,S'} \left[\max_{f \in \mathcal{H}_{\mathcal{C}}} |\hat{\mathcal{R}}_{S'}(f) - \hat{\mathcal{R}}_S(f)| \right] \\ &= \mathbb{E}_{S,S'} \left[\max_{f \in \mathcal{H}_{\mathcal{C}}} \left| \frac{1}{m} \sum_{i=1}^m \ell(f(x'_i), y'_i) - \ell(f(x_i), y_i) \right| \right] \end{aligned}$$



The VC Dimension

- Note the symmetry between S and S' : for any $\sigma \in \{\pm 1\}^m$,

$$\mathbb{E}_{S,S'} \left[\max_{f \in \mathcal{H}_C} \left| \frac{1}{m} \sum_{i=1}^m \ell(f(x'_i), y'_i) - \ell(f(x_i), y_i) \right| \right] = \mathbb{E}_{S,S'} \left[\max_{f \in \mathcal{H}_C} \left| \frac{1}{m} \sum_{i=1}^m \sigma_i (\ell(f(x'_i), y'_i) - \ell(f(x_i), y_i)) \right| \right]$$

- Thus, we can take expectation over a uniformly random $\sigma \sim \{\pm 1\}^n$:

$$\text{LHS} = \mathbb{E}_{S,S'} \mathbb{E}_{\sigma} \left[\max_{f \in \mathcal{H}_C} \left| \frac{1}{m} \sum_{i=1}^m \sigma_i (\ell(f(x'_i), y'_i) - \ell(f(x_i), y_i)) \right| \right]$$

- $\sigma_i (\ell(f(x'_i), y'_i) - \ell(f(x_i), y_i))$ is a mean-zero random variable in $\{-1, 0, 1\}$. By Hoeffding's inequality,

$$\Pr_{\sigma} \left[\left| \frac{1}{m} \sum_{i=1}^m \sigma_i (\ell(f(x'_i), y'_i) - \ell(f(x_i), y_i)) \right| \geq \epsilon \right] \leq \exp(-\Omega(m\epsilon^2))$$

The VC Dimension

- Note the symmetry between S and S' : for any $\sigma \in \{\pm 1\}^m$,

$$\mathbb{E}_{S,S'} \left[\max_{f \in \mathcal{H}_C} \left| \frac{1}{m} \sum_{i=1}^m \ell(f(x'_i), y'_i) - \ell(f(x_i), y_i) \right| \right] = \mathbb{E}_{S,S'} \left[\max_{f \in \mathcal{H}_C} \left| \frac{1}{m} \sum_{i=1}^m \sigma_i (\ell(f(x'_i), y'_i) - \ell(f(x_i), y_i)) \right| \right]$$

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The VC Dimension

- Using the fact that for any nonnegative random variable X ,

$$\mathbb{E}[X] = \int_0^\infty \Pr[X \geq t] dt,$$

we have

$$\begin{aligned} & \mathbb{E}_\sigma \left[\max_{f \in \mathcal{H}_C} \left| \frac{1}{m} \sum_{i=1}^m \sigma_i (\ell(f(x'_i), y'_i) - \ell(f(x_i), y_i)) \right| \right] \\ &= \int_0^\infty \Pr_\sigma \left[\max_{f \in \mathcal{H}_C} \left| \frac{1}{m} \sum_{i=1}^m \sigma_i (\ell(f(x'_i), y'_i) - \ell(f(x_i), y_i)) \right| \geq \epsilon \right] d\epsilon \\ &\leq \int_0^\infty \min\{1, \tau_{\mathcal{H}}(2m) \exp(-\Omega(m\epsilon^2))\} d\epsilon \\ &= O\left(\frac{\sqrt{1 + \log \tau_{\mathcal{H}}(2m)}}{\sqrt{m}}\right) \end{aligned}$$



The VC Dimension

$$\int_0^\infty \min\{1, \tau_H(2m) \exp(-\Omega(m\epsilon^2))\} d\epsilon$$

Worked for 1m 55s

Let

$$A = \tau_H(2m), \quad \exp(-\Omega(m\epsilon^2)) \leq e^{-cm\epsilon^2}$$

for some constant $c > 0$. Consider

$$I = \int_0^\infty \min\{1, Ae^{-cm\epsilon^2}\} d\epsilon.$$

The two terms inside the minimum are equal when

$$Ae^{-cm\epsilon_*^2} = 1 \implies \epsilon_* = \sqrt{\frac{\log A}{cm}}.$$

Thus, for $A \geq 1$,

$$\begin{aligned} I &= \int_0^{\epsilon_*} 1 d\epsilon + A \int_{\epsilon_*}^\infty e^{-cm\epsilon^2} d\epsilon \\ &= \sqrt{\frac{\log A}{cm}} + \frac{A\sqrt{\pi}}{2\sqrt{cm}} \operatorname{erfc}\left(\sqrt{\log A}\right). \end{aligned}$$

Using the Gaussian-tail bound

$$\int_x^\infty e^{-cm t^2} dt \leq \frac{e^{-cm x^2}}{2cm x},$$

we obtain

$$I \leq \sqrt{\frac{\log A}{cm}} + \frac{1}{2\sqrt{cm \log A}}.$$

Therefore, in a form valid even when A is close to 1,

$$\int_0^\infty \min\{1, \tau_H(2m)e^{-\Omega(m\epsilon^2)}\} d\epsilon = O\left(\sqrt{\frac{1 + \log \tau_H(2m)}{m}}\right).$$

The VC Dimension

Lemma (Sauer-Shelah-Perles). Let $\mathcal{H}$ be a hypothesis class with $\text{VCdim}(\mathcal{H}) \leq d < \infty$. Then, for all m ,

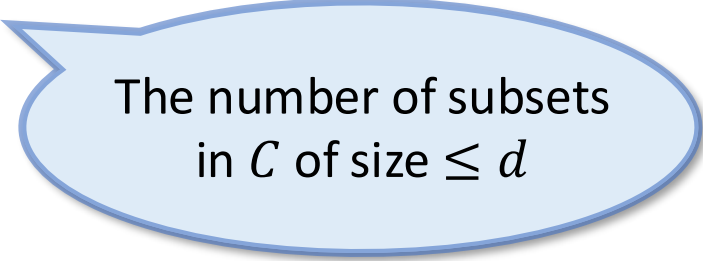
$$\tau_{\mathcal{H}}(m) := \max_{C \subset \mathcal{X} : |C| \leq m} |\mathcal{H}_C| \leq \sum_{i=0}^d \binom{m}{i}$$

Proof.

- We claim that for any $\mathcal{H}$ and $C \subset \mathcal{X}$,
$$|\mathcal{H}_C| \leq |\{B \subseteq C : \mathcal{H} \text{ shatters } B\}|$$
- We prove this claim by induction. Suppose it holds for $k < m$
- Fix $\mathcal{H}$ and $C = \{c_1, \dots, c_m\} \subset \mathcal{X}$
- Define two sets:

$$Y_0 = \{(y_2, \dots, y_m) : (0, y_2, \dots, y_m) \in \mathcal{H}_C \vee (1, y_2, \dots, y_m) \in \mathcal{H}_C\}$$

$$Y_1 = \{(y_2, \dots, y_m) : (0, y_2, \dots, y_m) \in \mathcal{H}_C \wedge (1, y_2, \dots, y_m) \in \mathcal{H}_C\}$$



The number of subsets
in C of size $\leq d$

The VC Dimension

- Define two sets:

$$Y_0 = \{(y_2, \dots, y_m) : (0, y_2, \dots, y_m) \in \mathcal{H}_C \vee (1, y_2, \dots, y_m) \in \mathcal{H}_C\}$$

$$Y_1 = \{(y_2, \dots, y_m) : (0, y_2, \dots, y_m) \in \mathcal{H}_C \wedge (1, y_2, \dots, y_m) \in \mathcal{H}_C\}$$

- $|Y_0| + |Y_1| = |\mathcal{H}_C|$

- Since $Y_0 = \mathcal{H}_{C'}$ with $C' := \{c_2, \dots, c_m\}$, we have

$$|Y_0| = |\mathcal{H}_{C'}| \leq |\{B \subseteq C' : \mathcal{H} \text{ shatters } B\}| = |\{B \subseteq C : c_1 \notin B \wedge \mathcal{H} \text{ shatters } B\}|$$

- Define $\mathcal{H}' \subset \mathcal{H}$ containing pairs of functions that only differ in c_1 :

$$\mathcal{H}' := \{h \in \mathcal{H} : \exists h' \in \mathcal{H} \text{ s.t. } h(c_i) = h'(c_i) \forall 2 \leq i \leq m \wedge 1 - h(c_1) = h'(c_1)\}$$

- Since $Y_1 = \mathcal{H}'_{C'}$, we have

$$\begin{aligned} |Y_1| &= |\mathcal{H}'_{C'}| \leq |\{B \subseteq C' : \mathcal{H}' \text{ shatters } B\}| = |\{B \subseteq C' : \mathcal{H}' \text{ shatters } B \cup \{c_1\}\}| \\ &= |\{B \subseteq C : c_1 \in B \wedge \mathcal{H}' \text{ shatters } B\}| \\ &\leq |\{B \subseteq C : c_1 \in B \wedge \mathcal{H} \text{ shatters } B\}| \end{aligned}$$

The VC Dimension

- Thus,

$$\begin{aligned} |\mathcal{H}_C| &= |Y_0| + |Y_1| \\ &\leq |\{B \subseteq C : c_1 \notin B \wedge \mathcal{H} \text{ shatters } B\}| + |\{B \subseteq C : c_1 \in B \wedge \mathcal{H} \text{ shatters } B\}| \\ &= |\{B \subseteq C : \mathcal{H} \text{ shatters } B\}| \end{aligned}$$



The VC Dimension

The Fundamental Theorem of Statistical Learning (quantitate version).

Let $\mathcal{H}$ be a hypothesis class of functions from $\mathcal{X}$ to $\{0, 1\}$ and let the loss function be the 0–1 loss. Suppose $\text{VCdim}(\mathcal{H}) = d$. Then

1) $\mathcal{H}$ is agnostic PAC learnable with sample complexity

$$m(\varepsilon, \delta) = \Theta\left(\frac{1 + \log(1/\delta)}{\varepsilon^2}\right)$$

2) $\mathcal{H}$ is PAC learnable with sample complexity

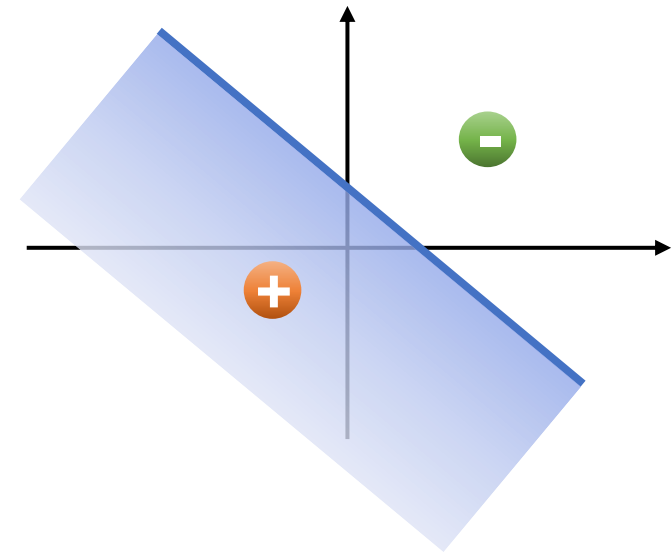
$$\Omega\left(\frac{d + \log(1/\delta)}{\varepsilon}\right) \leq m(\varepsilon, \delta) \leq O\left(\frac{d \log(1/\varepsilon) + \log(1/\delta)}{\varepsilon}\right)$$

Steve Hanneke (2016): $\mathcal{H}$ is **improperly** PAC learnable with sample complexity $\Theta\left(\frac{d + \log(1/\delta)}{\varepsilon}\right)$

The VC Dimension

The Halfspaces: $\mathcal{X} = \mathbb{R}^d$, $\mathcal{Y} = \{-1, 1\}$,

$$\mathcal{H} = \{\text{sgn}(\langle w, x \rangle + b) : w \in \mathbb{R}^d, b \in \mathbb{R}\}$$



Theorem.

- The VC dimension of the class of homogeneous halfspaces (i.e., $b = 0$) in $\mathbb{R}^d$ is d
- The VC dimension of the class of the nonhomogeneous halfspaces is $d + 1$

The Fundamental Theorem of Statistical Learning gives that $\mathcal{H}$ is PAC learnable with sample complexity

$$\frac{d + \log(1/\delta)}{\varepsilon}$$

The VC Dimension

Theorem. The VC dimension of the class of homogeneous halfspaces (i.e., $b = 0$) in $\mathbb{R}^d$ is d

Proof.

- It is easy to see that $\{e_1, \dots, e_d\}$ is shattered by $\mathcal{H}$: for any labeling $(y_1, \dots, y_d) \in \{\pm 1\}^d$, let
$$w := (y_1, \dots, y_d) \quad \Rightarrow \quad \langle w, e_i \rangle = y_i$$
- For any $\{x_1, \dots, x_{d+1}\} \subset \mathbb{R}^d$, suppose it can be shattered by $\mathcal{H}$
- There exist $c_1, \dots, c_{d+1}$ that are not all zero such that $\sum_{i=1}^{d+1} c_i x_i = 0$
- Let $I := \{i \in [d+1] : c_i > 0\}$ and $J := \{i \in [d+1] : c_i < 0\}$
- There are two cases:
 - I. Both I and J are non-empty
 - II. One of them I is empty

The VC Dimension

- Let $I := \{i \in [d + 1] : c_i > 0\}$ and $J := \{i \in [d + 1] : c_i < 0\}$
- If $I \neq \emptyset$ and $J \neq \emptyset$, then consider the labeling:

$$y_i := \text{sgn}(c_i)$$

- By shattering, there exists a $w \in \mathbb{R}^m$ such that $\text{sgn}(\langle w, x_i \rangle) = y_i$; that is,
$$\langle w, x_i \rangle \geq 0 \quad \text{if } i \in I, \quad \langle w, x_j \rangle < 0 \quad \text{if } j \in J$$

- Note that

$$\sum_{i \in I} c_i x_i = \sum_{j \in J} |c_j| x_j$$

- Then, we have

$$0 \leq \sum_{i \in I} c_i \langle w, x_i \rangle = \sum_{j \in J} |c_j| \langle w, x_j \rangle < 0$$

Contradiction!

The VC Dimension

- It remains to consider the case when $I \neq \emptyset$ and $J = \emptyset$; that is,

$$\sum_{i \in I} c_i x_i = 0, \quad c_i > 0 \quad \forall i \in I$$

- For any labeling with $y_i = -1$ for every $i \in I$, let w be the vector that realizes it, i.e.,

$$\langle w, x_i \rangle < 0 \quad \forall i \in I$$

- Then, we have

$$0 = \sum_{i \in I} c_i \langle w, x_i \rangle < 0$$

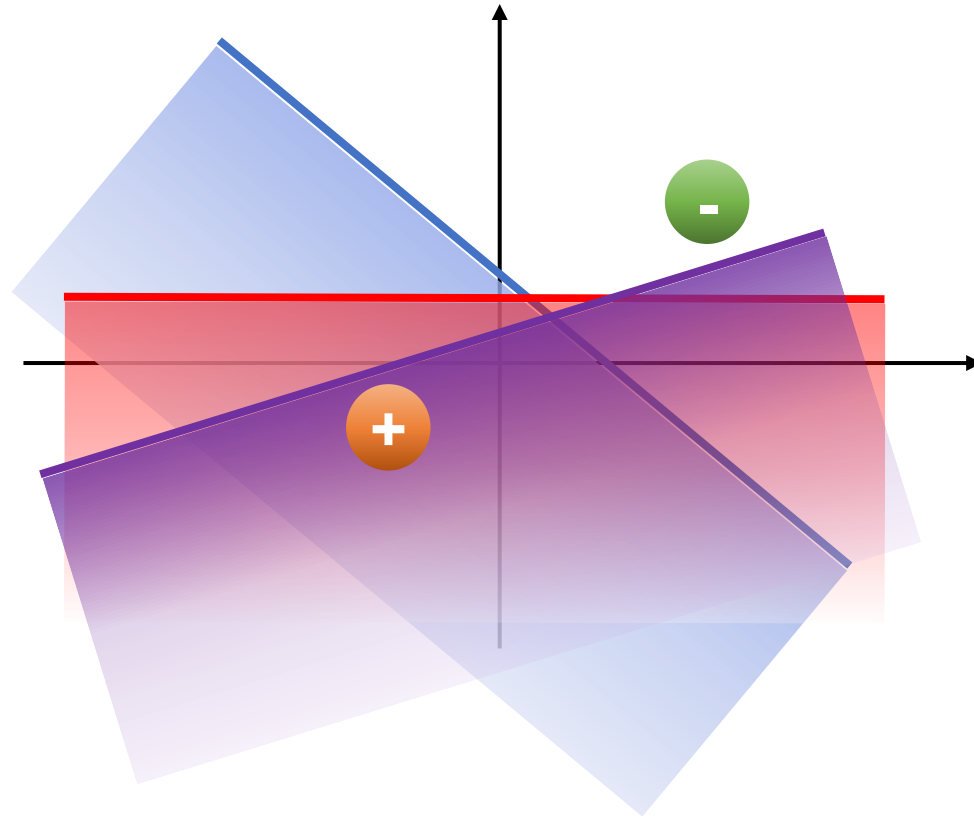
Contradiction!

- Therefore, no subset of size $d + 1$ can be shattered by $\mathcal{H}$
- Hence, $\text{VCdim}(\mathcal{H}) = d$



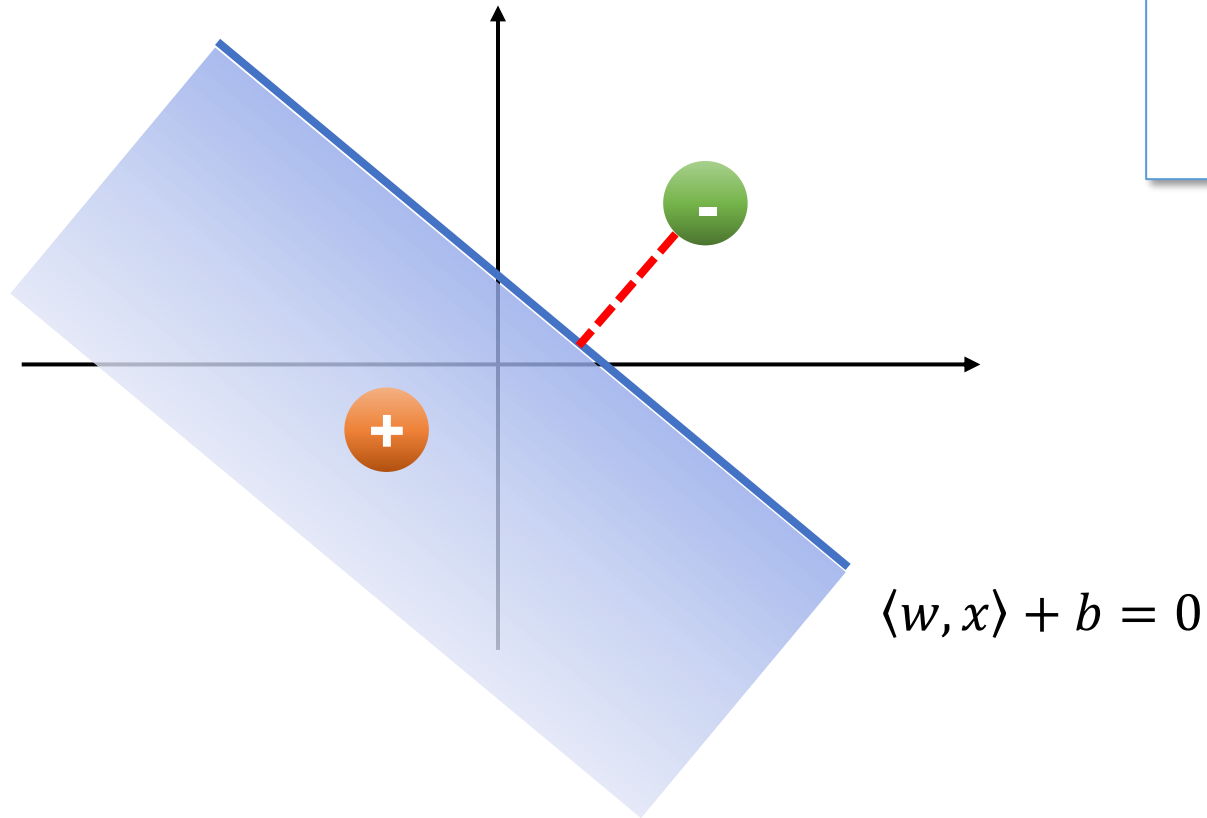
Support Vector Machine

Which halfspace is better?



Support Vector Machine

- Distance to the hyperplane



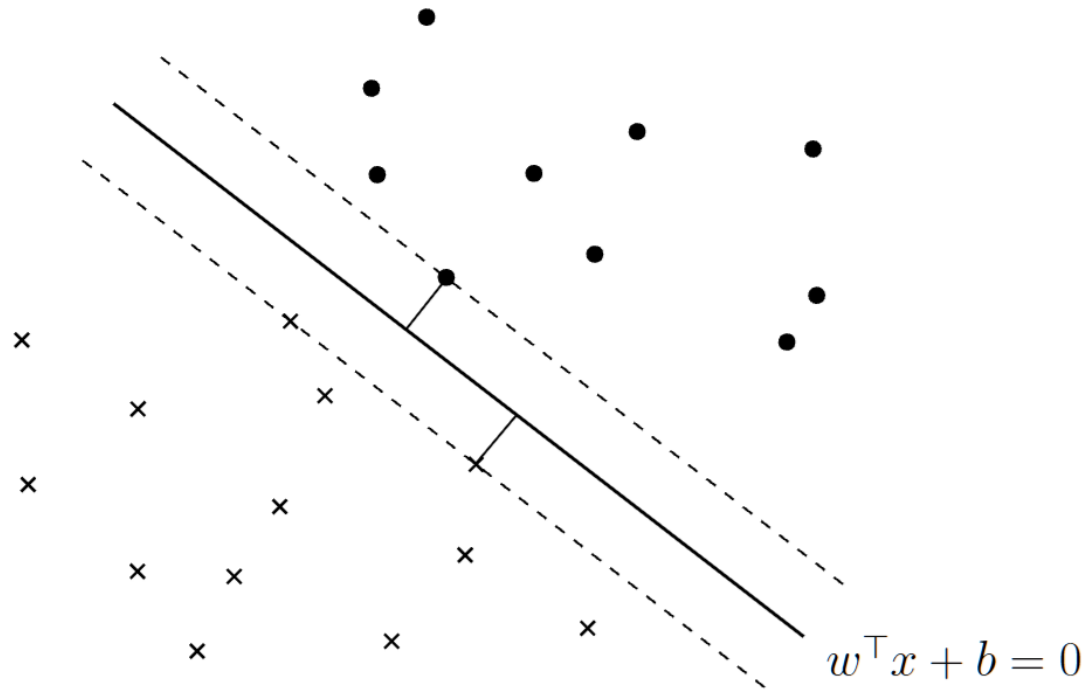
Geometric fact: The distance between a point x to the hyperplane (w, b) is

$$\frac{|\langle w, x \rangle + b|}{\|w\|}$$

Support Vector Machine

- **Margin:** the **minimal distance** between a point in the training set and the hyperplane

$$\min_{i \in [m]} \frac{|\langle w, x_i \rangle + b|}{\|w\|}$$



Support Vector Machine

- **Hard-SVM:** return an ERM hyperplane that separates the training set with the largest possible margin

$$\operatorname{argmax}_{(w,b): \|w\|=1} \min_{i \in [m]} |\langle w, x_i \rangle + b| \quad \text{s.t.} \quad y_i(\langle w, x_i \rangle + b) > 0 \quad \forall i \in [m] \quad (1)$$

- Whenever (1) is feasible (i.e., **linearly separable**), it is equivalent to the following optimization:

$$\operatorname{argmax}_{(w,b): \|w\|=1} \min_{i \in [m]} y_i(\langle w, x_i \rangle + b) \quad (2)$$

- It is also equivalent to the following quadratic optimization:

$$\begin{aligned} (w_0, b_0) &\leftarrow \operatorname{argmin}_{(w,b)} \|w\|^2 \quad \text{s.t.} \quad y_i(\langle w, x_i \rangle + b) \geq 1 \quad \forall i \in [m] \\ (w_*, b_*) &\leftarrow \left(\frac{w_0}{\|w_0\|}, \frac{b}{\|w_0\|} \right) \end{aligned} \quad (3)$$

Support Vector Machine

Duality:

$$\operatorname{argmin}_w \|w\|^2 \quad \text{s.t.} \quad y_i \langle w, x_i \rangle \geq 1 \quad \forall i \in [m]$$

- Define a penalty function

$$g(w) := \max_{\alpha \in \mathbb{R}^m: \alpha \geq 0} \sum_{i \in [m]} \alpha_i (1 - y_i (\langle w, x_i \rangle + b)) = \begin{cases} 0 & \text{if } y_i \langle w, x_i \rangle \geq 1 \quad \forall i \in [m] \\ \infty & \text{otherwise} \end{cases}$$

- Then, we have

$$\begin{aligned} \min_w \frac{1}{2} \|w\|^2 + g(w) &= \min_w \max_{\alpha \in \mathbb{R}^m: \alpha \geq 0} \frac{1}{2} \|w\|^2 + \sum_{i \in [m]} \alpha_i (1 - y_i (\langle w, x_i \rangle + b)) \\ &= \max_{\alpha \in \mathbb{R}^m: \alpha \geq 0} \left[\min_w \frac{1}{2} \|w\|^2 + \sum_{i \in [m]} \alpha_i (1 - y_i (\langle w, x_i \rangle + b)) \right] \end{aligned}$$

Slater's
condition

- Thus, $w_\star = \sum_{i \in [m]} \alpha_i y_i x_i$ in the span of $\{x_1, \dots, x_m\}$

Support Vector Machine

Duality:

Primal:
$$\min_w \frac{1}{2} \|w\|^2 \quad \text{s. t.} \quad y_i \langle w, x_i \rangle \geq 1 \quad \forall i \in [m]$$

Dual:
$$\max_{\alpha \in \mathbb{R}^m: \alpha \geq 0} \sum_{i \in [m]} \alpha_i - \frac{1}{2} \left\| \sum_{i \in [m]} \alpha_i y_i x_i \right\|^2$$

- Suppose w^* is the optimal primal solution and α^* is the optimal dual solution
- **Strong duality** implies that

$$\frac{1}{2} \|w^*\|^2 = \sum_{i \in [m]} \alpha_i^* - \frac{1}{2} \left\| \sum_{i \in [m]} \alpha_i^* y_i x_i \right\|^2$$

Support Vector Machine

Duality:

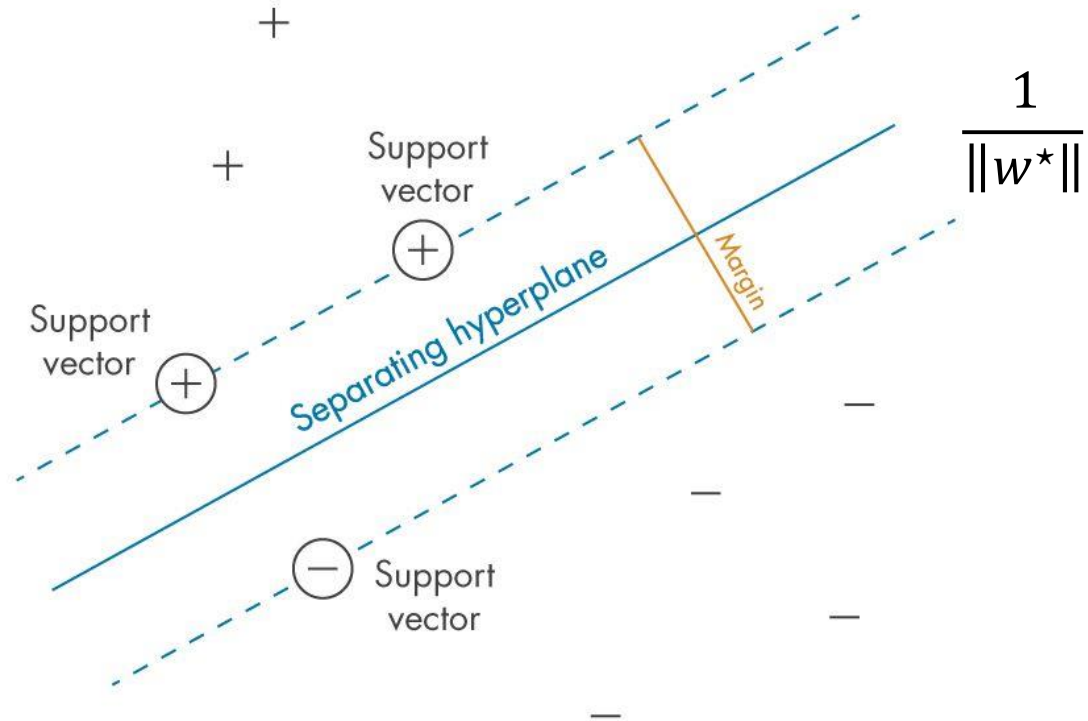
Primal:
$$\min_w \frac{1}{2} \|w\|^2 \quad \text{s. t.} \quad y_i \langle w, x_i \rangle \geq 1 \quad \forall i \in [m]$$

Dual:
$$\max_{\alpha \in \mathbb{R}^m: \alpha \geq 0} \sum_{i \in [m]} \alpha_i - \frac{1}{2} \left\| \sum_{i \in [m]} \alpha_i y_i x_i \right\|^2$$

- Suppose w^* is the optimal primal solution and α^* is the optimal dual solution
- **Strong duality** implies that

$$0 = \frac{1}{2} \left\| w^* - \sum_{i \in [m]} \alpha_i^* y_i x_i \right\|^2 + \sum_{i \in [m]} \alpha_i^* (y_i \langle w^*, x_i \rangle - 1)$$
$$w^* = \sum_{i \in [m]} \alpha_i^* y_i x_i, \quad \alpha_i^* (y_i \langle w^*, x_i \rangle - 1) = 0 \quad \forall i \in [m]$$

Support Vector Machine



What is the sample complexity of hard-SVM?

The Rademacher Complexity

- Let $\mathcal{F} := \ell \circ \mathcal{H} = \{(x, y) \mapsto \ell(h(x), y) : h \in \mathcal{H}\}$. Then, for any $f \in \mathcal{F}$, we have

$$\mathcal{R}(f) = \mathbb{E}_{(x,y) \sim \mathcal{D}}[f(x, y)], \quad \hat{\mathcal{R}}_S(f) = \frac{1}{m} \sum_{i=1}^m f(x_i, y_i)$$

- Representativeness of S w.r.t. $\mathcal{F}$:** the largest gap between the true error of a function f and its empirical error

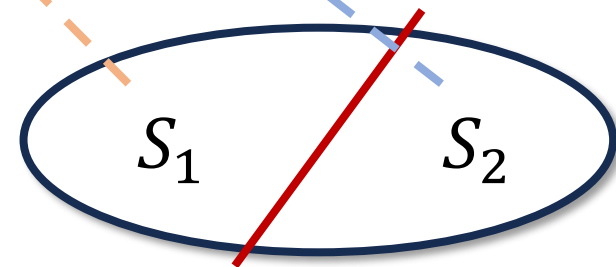
$$\text{Rep}_{\mathcal{D}}(\mathcal{F}, S) := \sup_{f \in \mathcal{F}} |\mathcal{R}(f) - \hat{\mathcal{R}}_S(f)|$$

- If $\text{Rep}_{\mathcal{D}}(\mathcal{F}, S) \leq \varepsilon/2$, then

$$\begin{aligned} \mathcal{R}(\text{ERM}_{\mathcal{H}}(S)) - \inf_{h \in \mathcal{H}} \mathcal{R}(h) &= \left(\mathcal{R}(\text{ERM}_{\mathcal{H}}(S)) - \hat{\mathcal{R}}_S(\text{ERM}_{\mathcal{H}}(S)) \right) + \left(\hat{\mathcal{R}}_S(\text{ERM}_{\mathcal{H}}(S)) - \hat{\mathcal{R}}_S(h) \right) \\ &\quad + \left(\hat{\mathcal{R}}_S(h) - \mathcal{R}(h) \right) \leq \varepsilon \end{aligned}$$

- How to estimate $\text{Rep}_{\mathcal{D}}(\mathcal{F}, S)$ using samples in S only?

$$\sup_{f \in \mathcal{F}} \left| \frac{1}{|S_1|} \sum_{i \in S_1} f(x_i, y_i) - \frac{1}{|S_2|} \sum_{j \in S_2} f(x_j, y_j) \right|$$



The Rademacher Complexity

$$\widehat{\text{Rep}}_{\mathcal{D}}(\mathcal{F}, S) := \sup_{f \in \mathcal{F}} \left| \frac{1}{|S_1|} \sum_{i \in S_1} f(x_i, y_i) - \frac{1}{|S_2|} \sum_{j \in S_2} f(x_j, y_j) \right|$$

- Suppose $|S_1| = |S_2| = m/2$. And let $\sigma \in \{-1, 1\}^m$ be such that
$$\sigma_i = 1 \quad \text{if } i \in S_1, \quad \sigma_i = -1 \quad \text{if } i \in S_2$$

$$\widehat{\text{Rep}}_{\mathcal{D}}(\mathcal{F}, S) = \frac{m}{2} \cdot \sup_{f \in \mathcal{F}} \left| \sum_{i=1}^m \sigma_i f(x_i, y_i) \right|$$

- The Rademacher complexity generalizes this idea by considering the expectation with respect to a random $\sigma \sim \{-1, 1\}^m$

The Rademacher Complexity

- Let $\mathcal{F} \circ S := \{(f(x_1, y_1), \dots, f(x_m, y_m)) : f \in \mathcal{F}\} \subset \mathbb{R}^m$
- Let $\sigma \in \{-1, 1\}^m$ be a random vector such that each coordinate is an independent Rademacher random variable:

$$\Pr[\sigma_i = \pm 1] = 1/2$$

- The **Rademacher complexity** of $\mathcal{F}$ with respect to S is defined as:

$$R(\mathcal{F} \circ S) := \frac{1}{m} \mathbb{E}_{\sigma} \left[\sup_{f \in \mathcal{F}} \sum_{i=1}^m \sigma_i f(x_i, y_i) \right] = \frac{1}{m} \mathbb{E}_{\sigma} \left[\sup_{v \in \mathcal{F} \circ S} \langle \sigma, v \rangle \right]$$

Lemma.

$$\mathbb{E}_{S \sim \mathcal{D}^m} [\text{Rep}_{\mathcal{D}}(\mathcal{F}, S)] \leq 2 \mathbb{E}_{S \sim \mathcal{D}^m} [R(\mathcal{F} \circ S)]$$

- The proof uses the symmetrization technique
- See (Shalev-Shwartz-Ben-David, Sec. 26.1) for more details

The Rademacher Complexity

Lemma.

$$\mathbb{E}_{S \sim \mathcal{D}^m} [\text{Rep}_{\mathcal{D}}(\mathcal{F}, S)] \leq 2 \mathbb{E}_{S \sim \mathcal{D}^m} [\mathcal{R}(\mathcal{F} \circ S)]$$

Corollary. For any $h \in \mathcal{H}$,

$$\mathbb{E}_{S \sim \mathcal{D}^m} [\mathcal{R}(\text{ERM}_{\mathcal{H}}(S)) - \mathcal{R}(h)] \leq 2 \mathbb{E}_{S \sim \mathcal{D}^m} [\mathcal{R}(\ell \circ \mathcal{H} \circ S)]$$

Proof.

- By the definition of $\text{Rep}_{\mathcal{D}}(\mathcal{F}, S)$, we have

$$\mathbb{E}_{S \sim \mathcal{D}^m} [\mathcal{R}(\text{ERM}_{\mathcal{H}}(S)) - \hat{\mathcal{R}}_S(\text{ERM}_{\mathcal{H}}(S))] \leq 2 \mathbb{E}_{S \sim \mathcal{D}^m} [\mathcal{R}(\ell \circ \mathcal{H} \circ S)]$$

- For any fixed $h \in \mathcal{H}$,

$$\mathcal{R}(h) = \mathbb{E}_{S \sim \mathcal{D}^m} [\hat{\mathcal{R}}_S(h)] \geq \mathbb{E}_{S \sim \mathcal{D}^m} [\hat{\mathcal{R}}_S(\text{ERM}_{\mathcal{H}}(S))]$$

How to convert this expectation bound to a high-probability bound? ■

Detour: Concentration (II)

$$\mathbb{E}[X] \leq ?? \quad \Rightarrow \quad \Pr[X \geq ???] \leq \delta$$

General approach: Markov inequality

For any **nonnegative** random variable X ,

$$\Pr\left[X \geq \frac{\mathbb{E}[X]}{\delta}\right] \leq \delta$$

- We have with probability $1 - \delta$ over $S \sim \mathcal{D}^m$,

$$\mathcal{R}(\text{ERM}_{\mathcal{H}}(S)) - \inf_{h \in \mathcal{H}} \mathcal{R}(h) \leq \frac{2\mathbb{E}_{S' \sim \mathcal{D}^m}[\mathcal{R}(\ell \circ \mathcal{H} \circ S')]}{\delta}$$

Detour: Concentration (II)

$$\mathbb{E}[X] \leq ?? \quad \Rightarrow \quad \Pr[X \geq ???] \leq \delta$$

Specialized approach: concentration inequalities

- If the random variable X has a specific tail condition (e.g., subgaussian tail), then we have

$$\Pr[X \geq \mathbb{E}[X] + c] \leq \Pr[|X - \mathbb{E}[X]| \geq c] \leq e^{-\mathcal{I}(c)}$$

- That is,

$$\Pr[X \geq \mathbb{E}[X] + \mathcal{I}^{-1}(\log 1/\delta)] \leq \delta$$

Example:

- Hoeffding's inequality: for any **fixed** $h \in \mathcal{H}$, with probability $1 - \delta$,

$$|\mathcal{R}(h) - \hat{\mathcal{R}}_S(h)| \lesssim \sqrt{\frac{\log(1/\delta)}{m}}$$

Detour: Concentration (II)

$$\mathbb{E}[X] \leq ?? \quad \Rightarrow \quad \Pr[X \geq ???] \leq \delta$$

Specialized approach: concentration inequalities

McDiarmid's Inequality

Let $\mathbb{V}$ be some set and let $f : \mathbb{V}^m \rightarrow \mathbb{R}$ be a function that satisfies the **bounded differences condition**: for all $i \in [m]$ and for all $x_1, \dots, x_m, x'_i \in \mathbb{V}$, we have

$$|f(x_1, \dots, x_m) - f(x_1, \dots, x_{i-1}, x'_i, x_{i+1}, \dots, x_m)| \leq c$$

Let $X_1, \dots, X_m$ be independent random variables supported on $\mathbb{V}$. Then, with probability $1 - \delta$,

$$|f(X_1, \dots, X_m) - \mathbb{E}[f(X_1, \dots, X_m)]| \leq O\left(c\sqrt{m \log(1/\delta)}\right)$$

The Rademacher Complexity

$$\text{Rep}_{\mathcal{D}}(\mathcal{F}, S) := \sup_{f \in \mathcal{F}} |\mathcal{R}(f) - \hat{\mathcal{R}}_S(f)| = \sup_{f \in \mathcal{F}} \left| \mathcal{R}(f) - \frac{1}{m} \sum_{i=1}^m f(x_i, y_i) \right|$$

- Suppose that the loss function is bounded, i.e., $|\ell(y, z)| \leq 1$
- Then, $\text{Rep}_{\mathcal{D}}(\mathcal{F}, S)$ satisfies the bounded differences condition with parameter $\frac{2}{m}$
- Recall that $\mathbb{E}_{S \sim \mathcal{D}^m}[\text{Rep}_{\mathcal{D}}(\mathcal{F}, S)] \leq 2\mathbb{E}_{S \sim \mathcal{D}^m}[\mathcal{R}(\mathcal{F} \circ S)]$
- McDiarmid's inequality gives that, with probability $1 - \delta$,

$$\begin{aligned} \text{Rep}_{\mathcal{D}}(\mathcal{F}, S) &\leq \mathbb{E}[\text{Rep}_{\mathcal{D}}(\mathcal{F}, S)] + O\left(\frac{2}{m} \sqrt{m \log(1/\delta)}\right) \\ &\leq O\left(\mathbb{E}_{S' \sim \mathcal{D}^m}[\mathcal{R}(\mathcal{F} \circ S')] + \sqrt{\frac{\log(1/\delta)}{m}}\right) \end{aligned}$$

The Rademacher Complexity

Theorem. Assume that the loss function ℓ is bounded by 1. Then, with probability $1 - \delta$,

$$\mathcal{R}(\text{ERM}_{\mathcal{H}}(S)) - \inf_{h \in \mathcal{H}} \mathcal{R}(h) \lesssim R(\ell \circ \mathcal{H} \circ S) + \sqrt{\frac{\log(1/\delta)}{m}}$$

Proof.

- We already show that with probability at least $1 - \delta$,

$$\mathcal{R}(\text{ERM}_{\mathcal{H}}(S)) - \hat{\mathcal{R}}_S(\text{ERM}_{\mathcal{H}}(S)) \leq \text{Rep}_{\mathcal{D}}(\mathcal{F}, S) \lesssim \mathbb{E}_{S' \sim \mathcal{D}^m} [R(\mathcal{F} \circ S')] + \sqrt{\frac{\log(1/\delta)}{m}}$$

- $R(\mathcal{F} \circ S') = \frac{1}{m} \mathbb{E}_{\sigma} \left[\sup_{f \in \mathcal{F}} \sum_{i=1}^m \sigma_i f(x_i, y_i) \right]$ also satisfies the bounded differences condition with parameter $\frac{2}{m}$

The Rademacher Complexity

Proof.

- We already show that with probability at least $1 - \delta/2$,

$$\text{Rep}_{\mathcal{D}}(\mathcal{F}, S) \lesssim \mathbb{E}_{S' \sim \mathcal{D}^m}[\text{R}(\mathcal{F} \circ S')] + \sqrt{\frac{\log(1/\delta)}{m}}$$

- By McDiarmid's inequality again, with probability $1 - \delta/2$,

$$|\text{R}(\mathcal{F} \circ S) - \mathbb{E}_{S' \sim \mathcal{D}^m}[\text{R}(\mathcal{F} \circ S')]| \lesssim \sqrt{\frac{\log(1/\delta)}{m}}$$

- Thus, by union bound, with probability $1 - \delta$,

$$\text{Rep}_{\mathcal{D}}(\mathcal{F}, S) \lesssim \text{R}(\ell \circ \mathcal{H} \circ S) + \sqrt{\frac{\log(1/\delta)}{m}}$$

The Rademacher Complexity

Proof.

- By the error decomposition:

$$\begin{aligned}\mathcal{R}(\text{ERM}_{\mathcal{H}}(S)) - \inf_{h \in \mathcal{H}} \mathcal{R}(h) &\leq \left(\mathcal{R}(\text{ERM}_{\mathcal{H}}(S)) - \hat{\mathcal{R}}_S(\text{ERM}_{\mathcal{H}}(S)) \right) + \left(\hat{\mathcal{R}}_S(h) - \mathcal{R}(h) \right) \\ &\leq 2\text{Rep}_{\mathcal{D}}(\mathcal{F}, S)\end{aligned}$$

- Thus, with probability at least $1 - \delta$,

$$\mathcal{R}(\text{ERM}_{\mathcal{H}}(S)) - \inf_{h \in \mathcal{H}} \mathcal{R}(h) \lesssim R(\ell \circ \mathcal{H} \circ S) + \sqrt{\frac{\log(1/\delta)}{m}}$$



The Rademacher Complexity

Properties of the Rademacher complexity

- If $A \subseteq A' \subseteq \mathbb{R}^m$, then $R(A) \leq R(A')$
- For any $A \subseteq \mathbb{R}^m$, $c \in \mathbb{R}$, and $a_0 \in \mathbb{R}^m$,
$$R(\{ca + a_0 : a \in A\}) = |c|R(A)$$
- Let $A \subseteq \mathbb{R}^m$ and let $A' = \{\sum_{i=1}^N c_i a_i : N \in \mathbb{N}, a_i \in A, c_i \geq 0, \|c\|_1 = 1\}$. Then, $R(A') = R(A)$

Massart lemma. Let $A = \{a_1, \dots, a_N\} \subset \mathbb{R}^m$. Define $\bar{a} := \frac{1}{N} \sum_{i=1}^N a_i$. Then,

$$R(A) \leq \max_{i \in [N]} \|a_i - \bar{a}\| \frac{\sqrt{2 \log N}}{m}$$

Contraction lemma. For each $i \in [m]$, let $\phi_i : \mathbb{R} \rightarrow \mathbb{R}$ be ρ -Lipschitz: for all $x, y \in \mathbb{R}$, we have $|\phi_i(x) - \phi_i(y)| \leq \rho|x - y|$. For $A \subset \mathbb{R}^m$, let $\phi \circ A := \{(\phi_1(a_1), \dots, \phi_m(a_m)) : a \in A\} \subset \mathbb{R}^m$. Then,
$$R(\phi \circ A) \leq \rho \cdot R(A)$$

The Rademacher Complexity

Lemma. Let $\mathcal{H}_2 := \{x \mapsto \langle w, x \rangle : \|w\| \leq 1\}$. For any $S = (x_1, \dots, x_m)$,

$$R(\mathcal{H}_2 \circ S) = R(\{(\langle w, x_1 \rangle, \dots, \langle w, x_m \rangle) : \|w\| \leq 1\}) \leq \frac{\max_i \|x_i\|}{\sqrt{m}}$$

Proof.

$$\begin{aligned} mR(\mathcal{H}_2 \circ S) &= \mathbb{E}_\sigma \left[\sup_{v \in \mathcal{H}_2 \circ S} \langle \sigma, v \rangle \right] = \mathbb{E}_\sigma \left[\sup_{\|w\| \leq 1} \sum_{i=1}^m \sigma_i \langle w, x_i \rangle \right] \\ &= \mathbb{E}_\sigma \left[\sup_{\|w\| \leq 1} \left\langle w, \sum_{i=1}^m \sigma_i x_i \right\rangle \right] = \mathbb{E}_\sigma \left[\left\| \sum_{i=1}^m \sigma_i x_i \right\| \right] \\ &\leq \left(\mathbb{E}_\sigma \left[\left\| \sum_{i=1}^m \sigma_i x_i \right\|^2 \right] \right)^{1/2} = \left(\sum_{i=1}^m \|x_i\|^2 \right)^{1/2} \\ &\leq \sqrt{m} \max_i \|x_i\| \end{aligned}$$

Jensen's
inequality



Sample Complexity of Hard-SVM

Theorem. Let $\mathcal{D}$ be a probability distribution over $\mathcal{X} \times \{\pm 1\}$ such that there exists w^* with

$$\Pr_{(x,y) \sim \mathcal{D}} [y_i \langle w^*, x_i \rangle \geq 1] = 1, \quad \text{and} \quad \|x\| \leq R$$

Let w_S be the solution of Hard-SVM:

$$\operatorname{argmin}_w \|w\|^2 \quad \text{s.t.} \quad y_i \langle w, x_i \rangle \geq 1 \quad \forall i \in [m]$$

Then, with probability at least $1 - \delta$ over $S \sim \mathcal{D}^m$,

$$\Pr_{(x,y) \sim \mathcal{D}} [y \neq \operatorname{sgn}(\langle w_S, x \rangle)] \leq O\left(\frac{R\|w^*\| + \log(1/\delta)}{\sqrt{m}}\right)$$

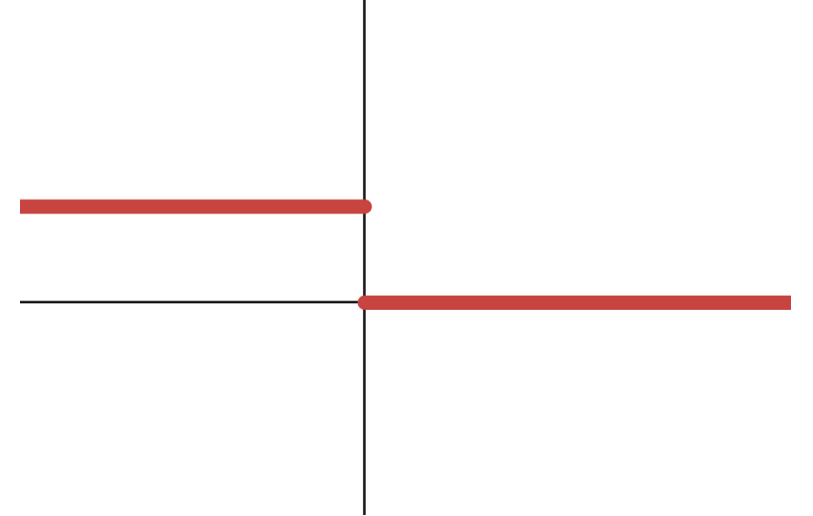
- The sample complexity is $O\left(\frac{R^2\|w^*\|^2}{\epsilon^2}\right)$, **without** any explicit dependence on the **dimension**
- The VC dimension of the class of halfspaces is d

Sample Complexity of Hard-SVM

- We know that with probability $1 - \delta$,

$$\mathcal{R}(h) - \hat{\mathcal{R}}_S(h) \lesssim R(\ell \circ \mathcal{H} \circ S) + \sqrt{\frac{\log(1/\delta)}{m}}$$

- $\Pr_{(x,y) \sim \mathcal{D}} [y \neq \text{sgn}(\langle w_S, x \rangle)] = \mathbb{E}[\ell_{0-1}(\text{sgn}(\langle w_S, x \rangle), y)]$
- However, the function $a \mapsto \ell_{0-1}(\text{sgn}(a), y)$ is not Lipschitz



Sample Complexity of Hard-SVM

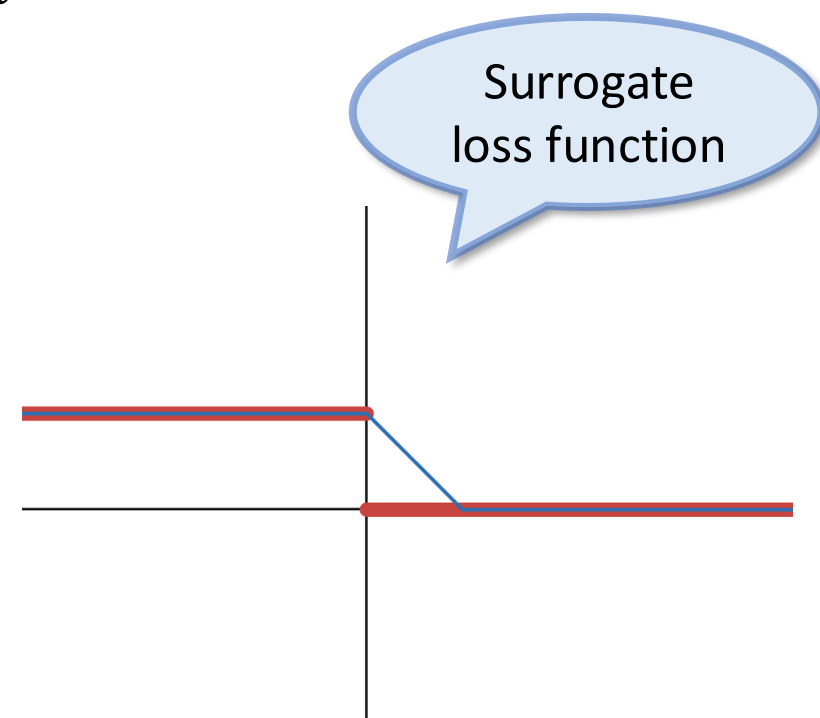
- We know that with probability $1 - \delta$,

$$\mathcal{R}(h) - \hat{\mathcal{R}}_S(h) \lesssim \mathcal{R}(\ell \circ \mathcal{H} \circ S) + \sqrt{\frac{\log(1/\delta)}{m}}$$

- $\Pr_{(x,y) \sim \mathcal{D}} [y \neq \text{sgn}(\langle w_S, x \rangle)] = \mathbb{E}[\ell_{0-1}(\text{sgn}(\langle w_S, x \rangle), y)]$
- However, the function $a \mapsto \ell_{0-1}(\text{sgn}(a), y)$ is not Lipschitz
- Define the **ramp loss**

$$a \mapsto \ell_{\text{ramp}}(a, y) := \min\{1, \max\{0, 1 - ya\}\}$$

- For $y \in \{\pm 1\}$, $\ell_{\text{ramp}}(\cdot, y)$ is **1-Lipschitz** and **bounded by 1**
- $\Pr_{(x,y) \sim \mathcal{D}} [y \neq \text{sgn}(\langle w, x \rangle)] \leq \mathbb{E}[\ell_{\text{ramp}}(\langle w, x \rangle, y)] = \mathcal{R}(h_w)$



Sample Complexity of Hard-SVM

- Let $\mathcal{H} := \{x \mapsto \langle w, x \rangle : \|w\| \leq \|w^*\|\}$
- Then, with probability 1, w_S (the Hard-SVM solution) is in $\mathcal{H}$ and $\hat{\mathcal{R}}_S(w_S) = 0$
- By Contraction lemma,

$$R(\ell_{\text{ramp}} \circ \mathcal{H} \circ S) \leq R(\mathcal{H} \circ S) = \|w^*\| \cdot R(\mathcal{H}_2 \circ S) \leq \|w^*\| \frac{\max_i \|x_i\|}{\sqrt{m}} \leq \frac{\|w^*\| R}{\sqrt{m}}$$

- Thus, with probability $1 - \delta$,

$$\Pr_{(x,y) \sim \mathcal{D}} [y \neq \text{sgn}(\langle w_S, x \rangle)] \leq O\left(\frac{\|w^*\| R + \sqrt{\log(1/\delta)}}{\sqrt{m}}\right)$$

